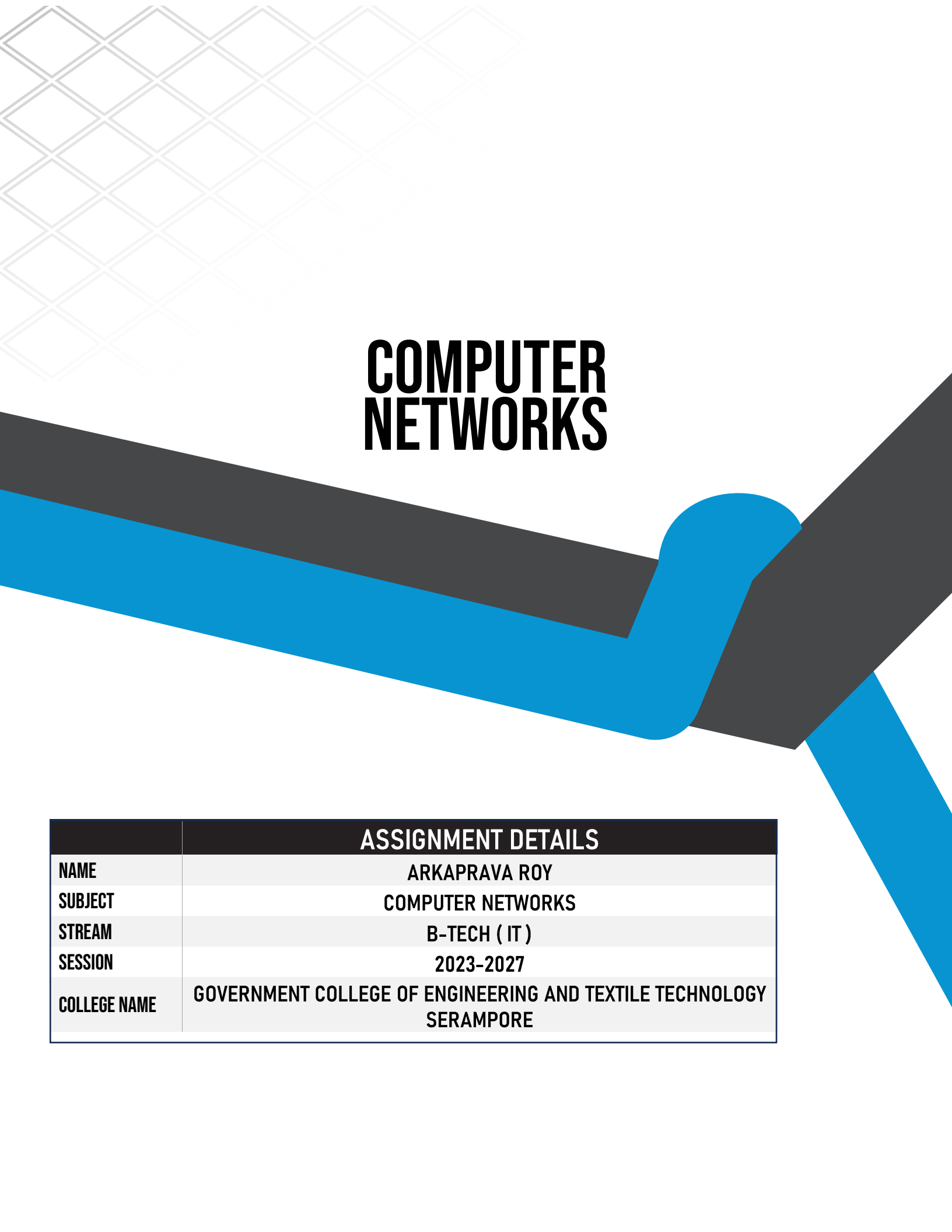
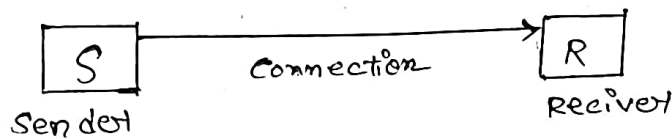




COMPUTER NETWORKS

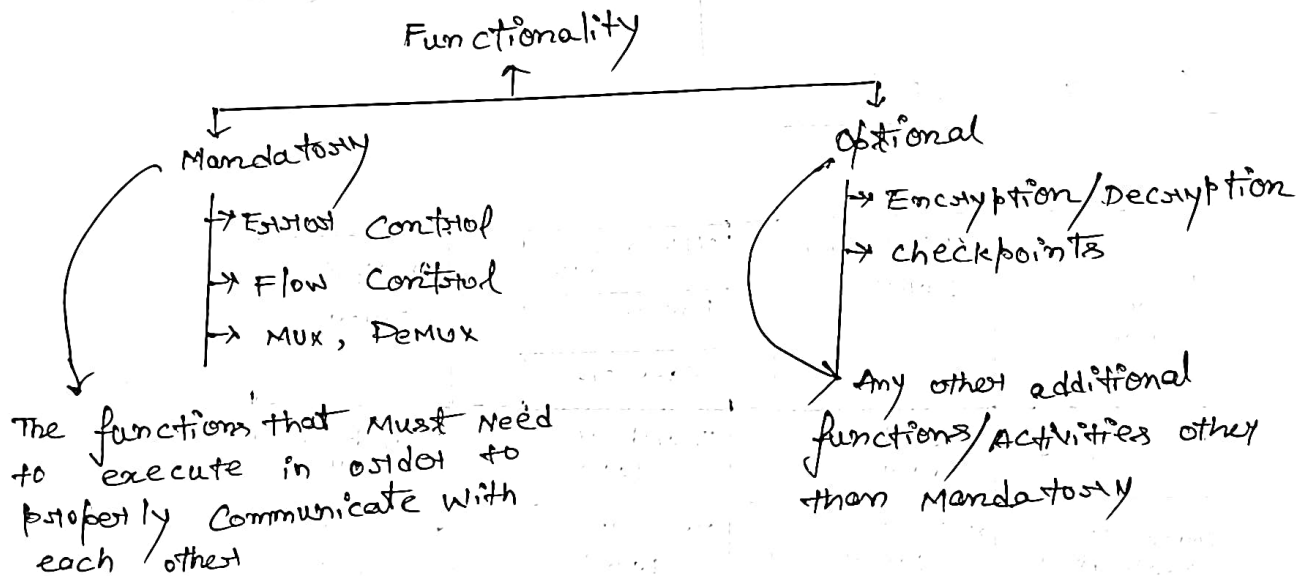
	ASSIGNMENT DETAILS
NAME	ARKAPRAVA ROY
SUBJECT	COMPUTER NETWORKS
STREAM	B-TECH (IT)
SESSION	2023-2027
COLLEGE NAME	GOVERNMENT COLLEGE OF ENGINEERING AND TEXTILE TECHNOLOGY SERAMPORE

INTRODUCTION TO NETWORKS



→ A standard protocol is used for communication for properly understood the sent/received message sent by sender/receiver

⊙ Sender & Receiver can be also known as client & server in different context.



⊙ OSI Model (Open System Interconnection Model) :

It is a 7-layer framework used in computer networks to explain how data travels from one device to another device over a network.

Top → Bottom

- I) Application Layer
- II) presentation Layer
- III) session Layer
- IV) Transport Layer

- V) Network Layer
- VI) Data Link Layer
- VII) Physical Layer

PAN, MAN, WAN, LAN, CAN

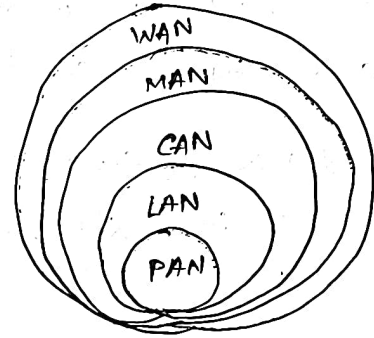
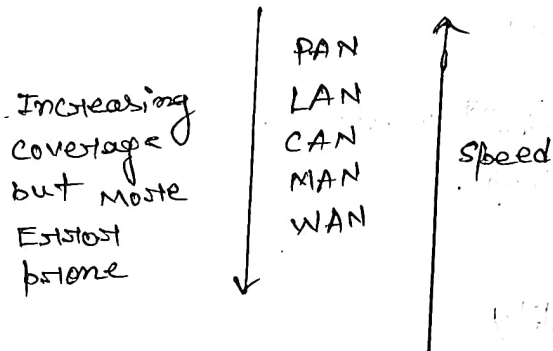
PAN → personal Area Network, Ex → Bluetooth, IrDA,

LAN → Local Area Network, Ex → Ethernet & WIFI

CAN → Campus Area Network, Ex → Ethernet

MAN → Metropolitan Area Network, Ex → FDDI, CDDI

WAN → wide Area Network, Ex → Dial-up



OSI vs TCP/IP

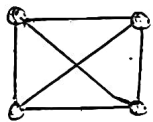
process to process	Application Layer	Application Layer Presentation Session	Application Layer
Host to Host	Transport Layer	Transport Layer	Transport Layer
Source to Destination	Network Layer	Network Layer	Internet Layer
Node to Node	Data Link Layer Physical Layer	Data Link Layer Physical Layer	Network Access Layer
	5 Layers TCP/IP	OSI	4-Layers

Physical Layer & its functions :

- Cables & Connectors
- physical Topology
- Hardware (Repeaters, Hubs)
- Transmission Mode
- Multiplexing
- Encoding

Types of Topology :

I) Mesh : Every system/points are connected with each other



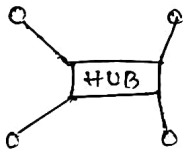
If, No of Device = n

Then, No of cable = $\frac{n(n-1)}{2}$

No of port $\rightarrow (n-1)n$

- ⊗ Reliable
- ⊗ Cost high
- ⊗ High maintenance
- ⊗ Highly secure.
- ⊗ point-to-point

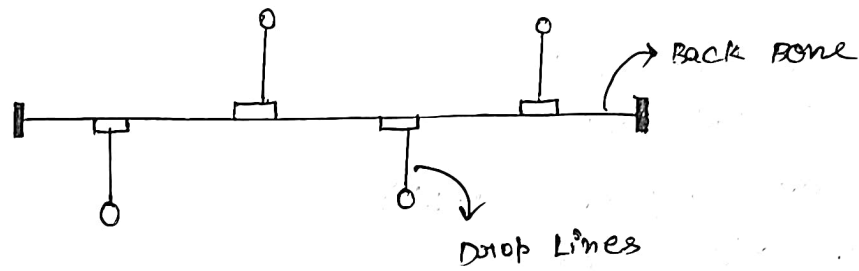
II) Hub / Star : All devices are connected with a common device. All signal goes through the common point/Hub.



No of port = $n \times 1 = n$

- ⊗ Not Reliable (If Hub fails, the entire system fails)
- ⊗ point-to-point
- ⊗ Expensive but less Expensive than Mesh.

III) BUS :

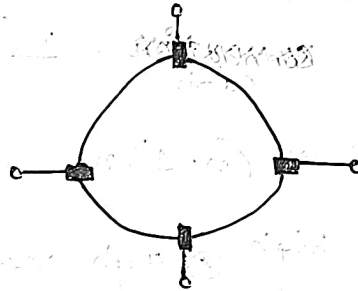


No of cable = $(n+1)$

No of port = n

- Ⓐ Not Reliable (single point failure)
- Ⓑ Less secure
- Ⓒ Not very costly

IV) Ring :



No of cable = $(n+1)$

No of port = (n)

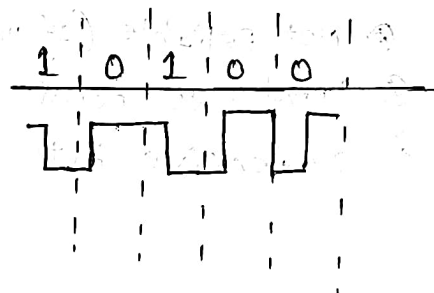
- Ⓐ Not Reliable
- Ⓑ Less secure
- Ⓒ cheap

Manchester Encoding

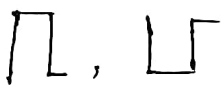
1 →

0 →

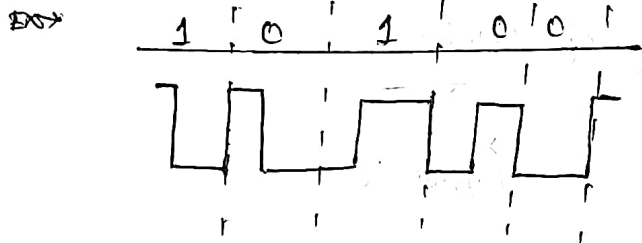
Ex →



Differential Manchester Encoding

0 → 

1 → 



Cables & Their Types

Cables

- Twisted pair Cable
- Coaxial Cable
- Fiber Optic Cable

Repeater

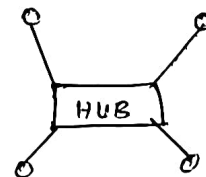


↓ Repeater

- Repeats the signal & helps it to travel further
- Doesn't Amplify the signal.

Hub

- Hub is a Multipoint Repeater
- Has no filters, it only forwards



Bridges

- Has filters
- Can forward Data,
- Bridges maintain a table of Mac & ports to function like a filter.

Two types of Bridge

static

- only can identify a device by its Mac address
- can't handle the situation if Mac is changed

dynamic

- Dynamic Bridges can learn about Mac and identifies devices
- can easily handle any situation if Mac changes. it learns from situation

- There is no collision on bridge, since it has its own buffer and it follows 'store & forward' policy.

Switch

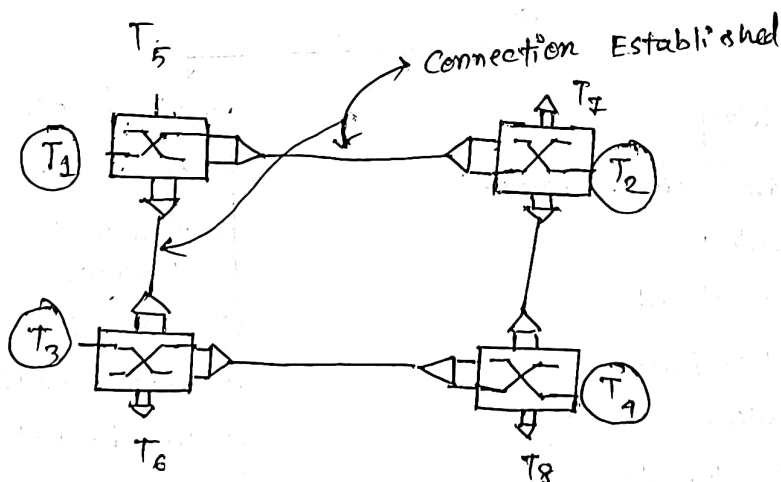
- Switch is a multipoint Bridge.
- It has a full duplex link. meaning a data can travel in both ways from sender to receiver without any collision
- Collision Domain is 0.
- Traffic is minimum.

Router

- Can forward Data
- Checks both Mac, and IP address
- Holds a Routing Table, which is used for sending Request to its destination.
- If Router can't understand where to send Data, it uses 'Flooding', Means, it sends/broadcast the data everywhere.
- It is a Multiport device.
- It has Collision Domain of 0, because it uses 'store & forward' policy.

Circuit Switching

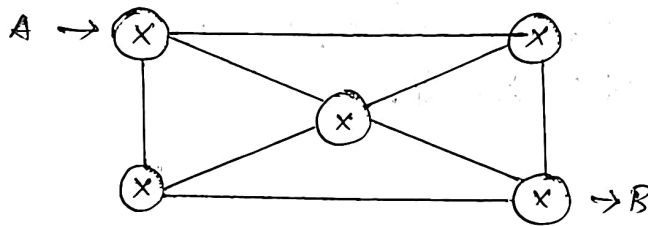
- Happens in Physical Layer
- Continuous flow
- Doesn't need Any header as it is a uninteruptive flow
- Resources are Reserved as Connection Established, so efficiency is Less
- Less Delay



$$\text{Total Time} = \text{Setup time} + \text{Transmission Time} \left(\frac{\text{Message}}{\text{Bandwidth}} \right) + \text{Propagation Delay} \left(\frac{\text{Distance}}{\text{Speed}} \right) + \text{Tear Down Time}$$

Packet Switching

- Happens in Data link & Network Layer
- store and forward
- pipelining used
- Efficient, as resources are not reserved
- Has a certain Delay



- Data is divided into packets
- packets are stored and then forwarded.
- for the 'storing' there is some delay

Total time → $n \times \text{Transmission Time} + \text{propagation Delay}$

↓
No of checkpoints / switches

Packet Switching

Datagram Switching

- No reservation of resource
- out of order
- High on overhead
- High packet loss
- Connectionless
- Low cost due to No reservation

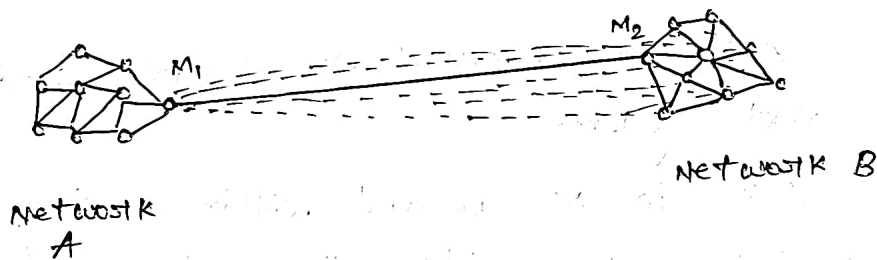
Virtual circuit

- Connection Oriented
- Reservation of Resource
- Same Order
- Less overhead
- Low packet loss
- High cost due to reservation

Message Switching

- Predecessor of packet switching
- store & forward Technique
- Message switching is just packet switching without dividing the data (message) into small sub divisions (packets)
- Hop by Hop (Checkpoint - to checkpoint)
- No concept of parallelism / pipeline as the data is sent as whole without dividing.

Unicast, Broadcasting & Multicasting



Unicast: If M_1 sends any message to M_2 , privately, that is called unicast

Broadcast: If M_1 sends the message to whole Network B, that is called Broadcast.

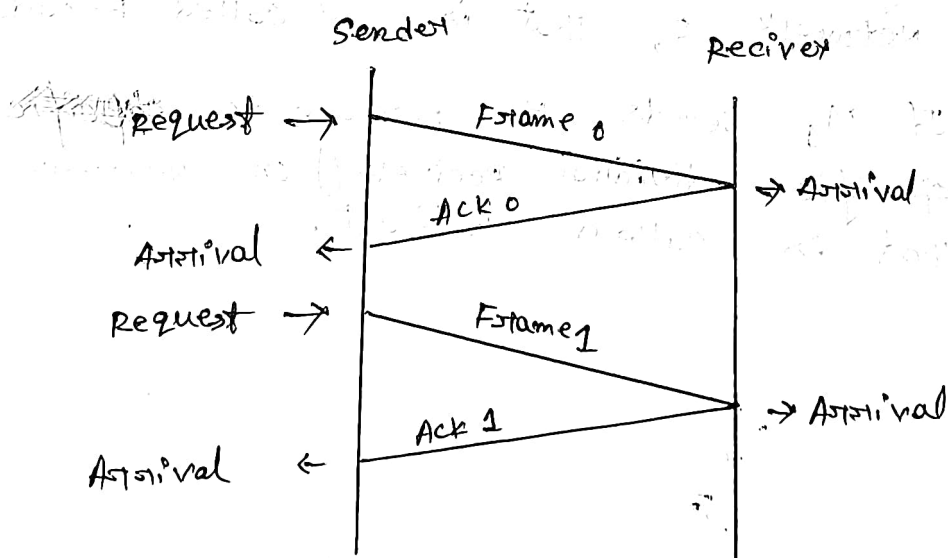
Multicast: If M_1 sends the message to specific individual receivers on Network B, that is called Multicast.

Data Link Layer

- Hop to Hop & Node to Node Delivery
- Flow Control
- Error Control (CRC, Check-sum)
- Access Control (CSMA/CD, Aloha, Token)
- Physical Address (MAC) (48 bit Address)

Stop and wait ARQ protocol

- Sender sends one frame (packet) and then stops
- Sender waits for an ACK (acknowledgement) from receiver.
- If ACK is received → Sender sends the next frame
- If ACK is not received within a timeout → Sender retransmits the same frame
- Frames carry a 1 bit sequence number (0 or 1) to detect duplicates.



Go Back N ARQ

- I) Sender can transmit N - frames continuously without waiting for ACKs.
- II) Each frame has a sequence No.
- III) The Receiver only accepts frames in order and rejects out of order frames.
- IV) Receiver sends cumulative ACKs.
- V) If a frame is lost/corrupted, the receiver stops accepting further frames and the sender sends the lost frame.

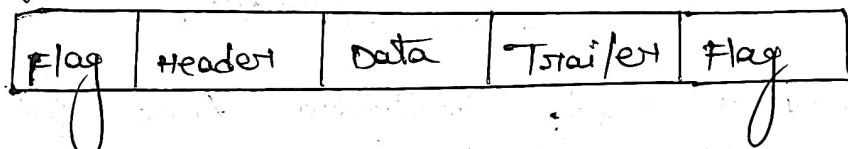
Selective Repeat ARQ

- I) Sender transmits multiple frames within a sliding window.
- II) Each frame has a sequence number.
- III) The Receiver: Accepts out-of-order frames, buffers them until missing frame arrives.
- IV) Receiver sends individual ACKs for each correctly received frame.
- V) Sender retransmits only the missing/corrupted frames after timeout.
- VI) More efficient than stop & wait and Go back N ARQ.

Framing

- Framing is needed because, the physical layer sends raw bits only.
- So the receiver must know, the start and the end of data.
- Sender receives a packet from the network layer.
- It adds a frame header and trailer
 - Header: Addressing, control info
 - Trailer: Error Detection (CRC)
- Frame is sent as a bit stream
- receiver detects frame boundaries
- Trailer is checked & error is detected
- Data is passed to the network layer

Structure:



Different Types of Framing

I) Character Count:

- Header contains number of character
- Receiver reads exactly that many characters
- If count field is corrupted, the entire frame breaks.

II) Byte Stuffing (Flag Byte)

→ special flag bytes marks the start & the end of frame.

→ If flag appears in data → extra escape byte is added.

III) Bit Stuffing

→ Frame begins and ends with a flag pattern:

Ex: 0111110

Flag pattern

→ After identifying flag pattern, the receiver removes the flags (0).

IV) Physical Layer Coding Violation

→ uses invalid signal patterns to mark frame boundaries.

→ works only if physical encoding allows unused signals

Error Detection & Control

Error

- Single bit Error (only one bit changes)
- Burst Error (more than one bit changes)

Detection

- simple parity (Even odd)
- 2D parity check
- check sum
- CRC

Correction

- Hamming code

Single parity Error Detection

- I) Sender Counts the Number of 1s in the Data
- II) A parity bit is added so that the total Number of 1s becomes:
 - Even
 - odd
- III) Receiver Counts 1s Again
- IV) If parity rule is broken → Error Detected

→ Simple, Low Overhead, Easy to implement
→ Detects only single bit error, and can't have any corrective measures

CRC

Sender Side

- I) Take Original Data
- II) Append extra 0s to the Data
(No. of 0s, = Generator length - 1)
- III) Divide using XOR division.
- IV) Get Remainder
- V) Replace appended 0s with Remainder
- VI) Send final Codeword.

Receiver Side

- I) Receive Codeword
- II) Divide by same Generator
- III) If Remainder = 0 → NO error

Ex → Data → 1101011011

Generator → 10011

Length of Generator → 5

So append $(5-1) = 4$ 0s to Data,

Now, Data → 11010110110000

Now, we perform XOR operation

11010110110000 ÷ 10011

Remainder → 1110

Now, we replace the added 0 with remainder,

So, Final transmitted Codeword

→ 11010110111110

Receiver verification →

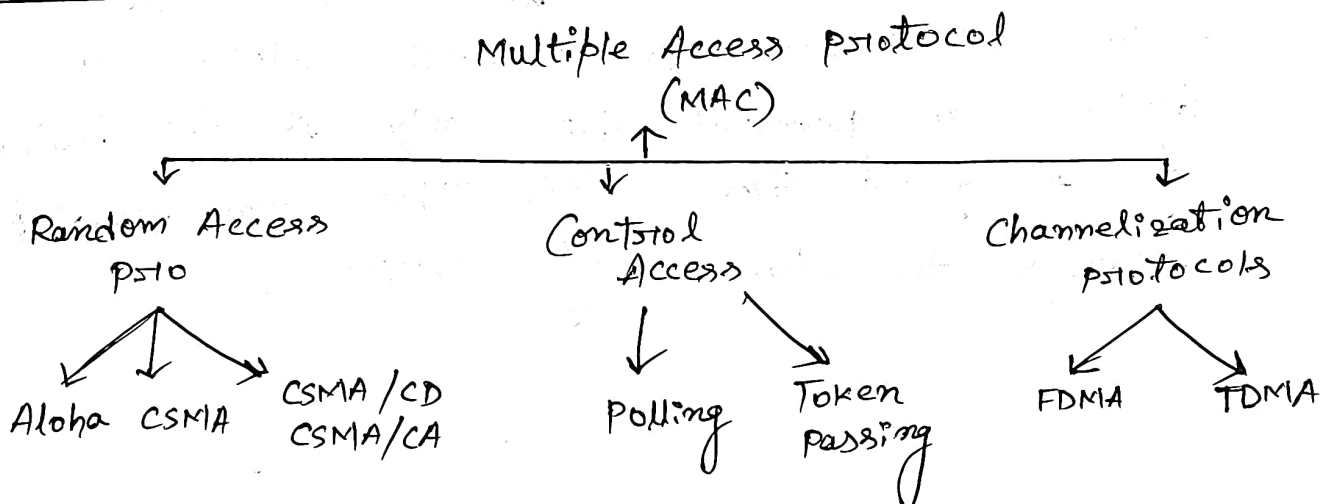
Receiver divides

11010110111110 ÷ 10011

If Remainder is → 0000

Then Data is Correct

Multiple Access protocol



Pure Aloha

Pure Aloha is one of the easiest and simplest MAC protocols. It's a Random access protocol where, a device sends data whenever it wants without checking whether the channel is busy or free. It is a LAN based protocol.

- ⇒ Station sends frame immediately
- ⇒ Receiver sends ACK if successful
- ⇒ If collision occurs, No ACK is received.
- ⇒ Sender waits random time
- ⇒ Retransmits frame.

$$\text{Transmission Time} = \frac{\text{Message Length}}{\text{Bandwidth}}$$

$$\text{Vulnerable Time} = (2 \times T_T)$$

$$\text{Efficiency } (\eta) = G \times e^{-2G} \quad G \rightarrow \text{number of stations}$$

Slotted Aloha

Slotted Aloha is an improved version of pure Aloha used in MAC protocols to reduce collisions. In slotted Aloha, a device can transmit data only at the beginning of a fixed time slot. This synchronization reduces the chance of collisions.

⇒ Time is divided into equal slots,

⇒ Now, a station, waits for next slot, then transmits.

⇒ If two devices transmits in same slot, then Collision Occurs. otherwise transmission successful

for slotted Aloha Vulnerable Time $= T = \frac{\text{Message len}}{\text{Bandwidth}}$

for pure Aloha it was $\rightarrow 2T$

So, Collision is reduced by half. Also efficiency is increased by 2x.

Let's calculate Efficiency of Both Pure Aloha and Slotted Aloha,

Pure Aloha

We know Efficiency $(\eta) = G \times e^{-2G}$

using differentiation,

$$\frac{dS}{dG} = e^{-2G} (1 - 2G)$$

for, Max throughput

$$\frac{dS}{dG} = 0$$

$$\text{So, } 1 - 2G = 0$$

$$\Rightarrow G = \frac{1}{2}$$

Then, Putting value,

$$\eta = \frac{1}{2} e^{-1}$$

$$\Rightarrow \eta = \frac{1}{2 \times 2.71} \approx 0.184 \quad (e = 2.71)$$
$$= 18.4 \%$$

Slotted Aloha

$$\text{Efficiency} = G e^{-G}$$

After Differentiating w.r. to G ,

$$\frac{dS}{dG} = e^{-G} (1 - G)$$

$$\text{Now, } 1 - G = 0$$

$$\Rightarrow G = 1$$

Then, putting value,

$$\eta = e^{-1}$$

$$= 2.71^{-1} = \frac{1}{2.71}$$

$$\approx 0.368$$

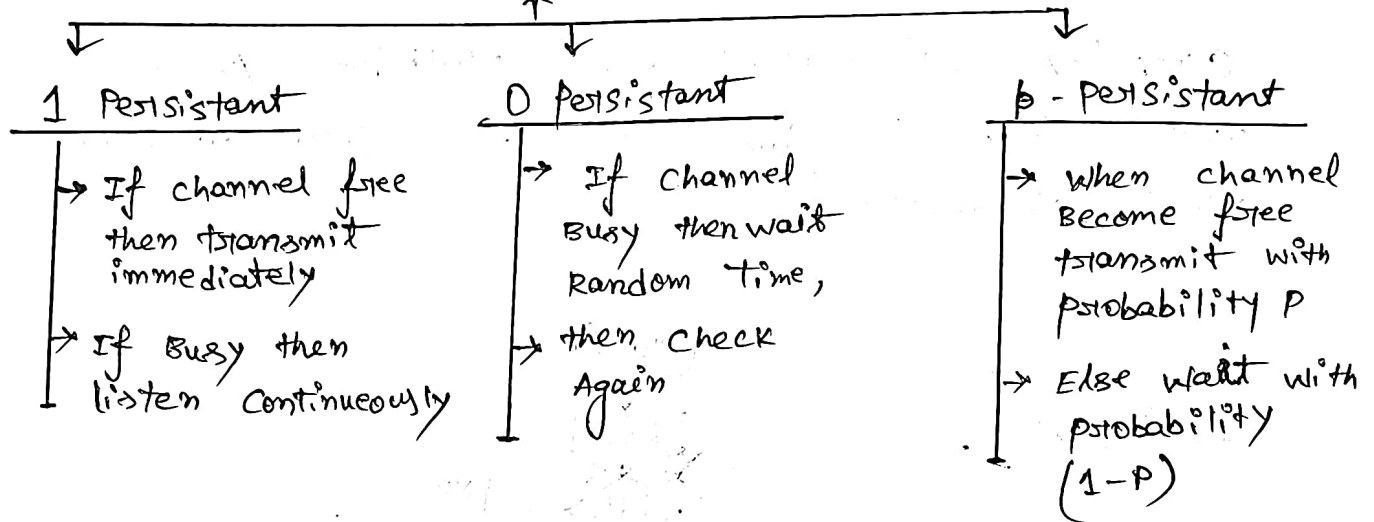
$$= 36.8\%$$

CSMA

CSMA (Carrier Sense Multiple Access) is a MAC protocol used in Computer Networks to reduce collisions. The main idea is 'Listen Before you talk'. Before sending the data a device first checks whether the communication channel is busy or free.

- ⇒ station wants to transmit.
- ⇒ It listens to the channel.
- ⇒ If channel free: Transmit immediately
- ⇒ If busy: wait
- ⇒ After channel become free: Transmit again.

CSMA



CSMA/CD

CSMA/CD or Carrier Sense Multiple Access with Collision Detection is used in traditional wired Ethernet.

- ⇒ Sense the channel :
 - If busy → wait
 - If free → transmit
- ⇒ Start Transmission
- ⇒ Detect Collision
- ⇒ Stop Transmission
- ⇒ Send Jam signal : Let all devices know that a collision happened.
- ⇒ Wait Random time

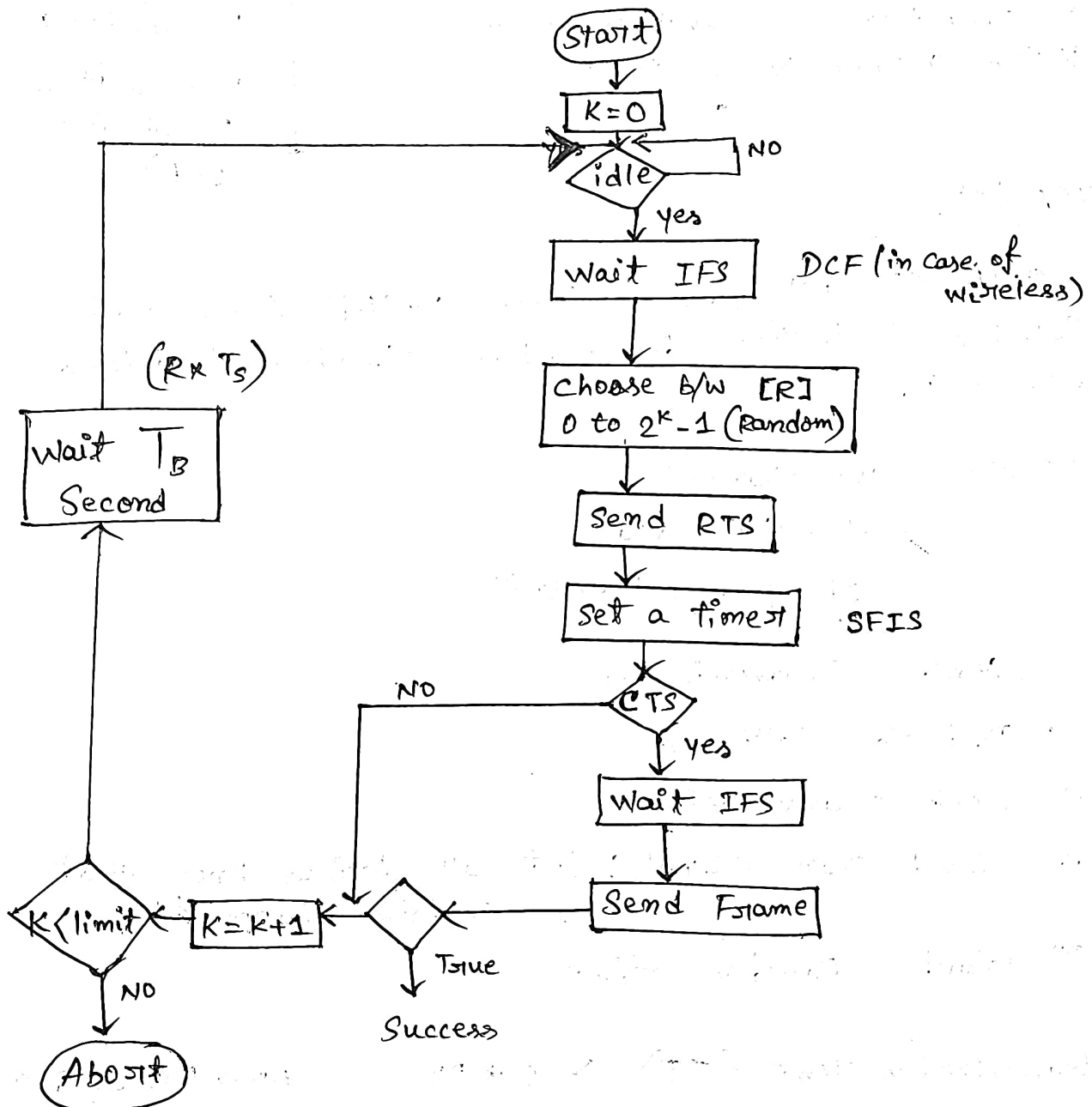
$$\text{Transmission Time (TT)} \geq 2 * PD$$

PD → Propagation Delay

$$\text{Efficiency } (\eta) = \frac{1}{1 + 6.44a} \quad \left(a = \frac{PD}{TT} \right)$$

CSMA / CA

CSMA / CA or Carrier Sense Multiple Access with Collision Avoidance is mainly used in Wifi Networks.



Ethernet Frame Format (IEEE 802.3)

Bus Topology	Preamble 7B	SFD 1B	DA 6B	SA 6B	Length 2B	Data 46B-1500B	CRC 4B
-----------------	----------------	-----------	----------	----------	--------------	-------------------	-----------

I) Preamble (7 byte) :

- Synchronizes sender's and receiver's clock
- Helps receiver prepare for incoming frame

II) Start Frame Delimiter (SFD) (1 byte) :

- Indicates actual frame starts now.

III) Destination Address (DA) (6 byte) :

- MAC Address of receiver

IV) Source Address (SA) (6 byte) :

- MAC Address of sender.

V) Length Field (2 byte) :

- Represents length of data

VI) Data Field (46 bits to 1500 bytes)

- Actual Payload / Data
- LLC information

VII) CRC/FCS (4 byte)

- Frame check sequence
- Error Detection using CRC.

Token Ring (IEEE 802.5)

- Ring Topology is used
- Access Control Method used in Token Passing.
- Token Ring is unidirectional
- Data rate used is, 4 Mbps & 16 Mbps
- Piggybacking Acknowledgement is used
- Differential Manchester Encoding is used.
- Variable Size Framing.
- Monitor station is used.

SD	AC	FC	DA	SA	Data	CRC	ED	FS	SD	AC	ED
1	1	1	6	6		(A)	1	1	1	1	1

SD → Start Delimiter

AC → Access Control

FC → Frame Control

DA → Destination Address

SA → Source Address

FCS → Frame Check Sequence

ED → End Delimiter

FS → Frame Status

Hamming Code

Hamming Code is an error-correcting code developed by Richard Hamming. used for:

- Detect Error
- Correct Single bit Error Automatically.

Ex $\rightarrow$ Suppose we want to send,

1011

Now, we need to add parity Bits.

S1: Determine The number of parity bits

formula: $2^r \geq (m+r+1)$ $m \rightarrow$ no of Data bits
 $r \rightarrow$ no of parity bits

For our Data, $m=4$

we try values,

$$r=2: 2^2 = 4, \text{ but } (4+2+1) = 7$$

$7 < 2^3$; so value not acceptable

$$\text{Now, } r=3: 2^3 = 8, \text{ and } (4+3+1) = 8$$

This value is acceptable.

So we need 3 parity bits.

S2: position The parity bits

parity bits are placed at position,

1, 2, 4, 8

So position becomes,

$P_1 \quad P_2 \quad D_1 \quad P_4 \quad D_2 \quad D_3 \quad D_4$

Inserting Data 1011:

$P_1 \quad P_2 \quad 1 \quad P_4 \quad 0 \quad 1 \quad 1$

S3: Calculate parity Bits

Let's assume Even parity.

calculate $P_1 \rightarrow$

P_1 checks position,
1, 3, 5, 7

Bits are, $P_1, 1, 0, 1$

Number of 1s = 2 (Already Even)

So,
 $P_1 = 0$

calculate $P_2 \rightarrow$

P_2 checks,
2, 3, 6, 7

Bits: $P_2, 1, 1, 1$

Numbers of 1s = 3 (odd)

We need one more 1 to make Even

So, $P_2 = 1$

calculate $P_4 \rightarrow$

P_4 checks,
4, 5, 6, 7

Bits: $P_4, 0, 1, 1$

Number of 1s = 2 (Even)

So, $P_4 = 0$

So, final Hamming Code: 0110011

S₄: Error Detection and Correction

Suppose receiver Gets: 0110001

One bit Changed during Transmission.

Now, Receiver recalculate parity bits. If Parity checks fail, it forms a binary number called Syndrome.

Ex[→]

$$P_4 \text{ error} = 1$$

$$P_2 \text{ error} = 1$$

$$P_1 \text{ error} = 0$$

$$\text{Binary: } 110_2 = 6$$

So, error is at position 6. Receiver flips bit 6 and Corrects the data automatically.

Sliding Window Protocol

Sliding window protocol is a flow control technique used in a Data link Layer and Transport Layer (TCP). It allows the Sender to send multiple frames before receiving acknowledgements, which improves network efficiency.

In stop and wait, sender sends one frame, waits for acknowledgement, and then sends another frame. The sender remains idle while waiting for Ack, which leads to poor utilization.

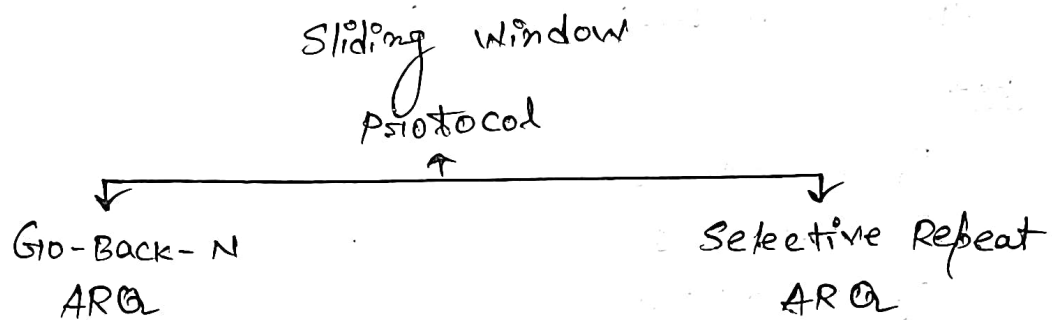
Working of Sliding Window →

Instead of sending one frame at a time,

→ Sender can send several frames continuously.

→ Receiver acknowledges them.

→ As Ack arrives, the window 'slides' forward.



Piggybacking

Piggybacking is a technique used in full duplex communication where an Acknowledgement is attached to a Data frame instead of sending a separate Ack frame.

Working

→ A sends a Data to B

→ B receives the Data

→ Instead of immediately sending Ack, B waits for a short time.

→ If B has data to send, it attaches the Ack to Data frame.

→ A extracts both Data and Ack.

Network Layer

The Network Layer is the third layer of the OSI Model. The main Goal of network layer, "Find the best path and deliver packets from Source Network to Destination Network."

Responsibilities of Network Layer

I) Logical Addressing:

Network Layer provides IP Address. This works as a logical Address in the Network Layer. Every device Gets 1. This helps to identify Source and Destination.

II) Routing:

Routing means, selecting the best path for packet transmission. Router uses routing algorithm and routing table for efficient routing.

III) Packet forwarding:

After route selection the router forwards packets towards destination. This is called forwarding.

IV) Inter networking:

Network Layer connects, different networks together, like LAN to WAN, wifi to internet etc.

V) Fragmentation and Reassembly :

Different networks support different packet size. If packet is too large, network layer breaks it into fragments. At destination fragments are reassembled.

- VI) Congestion Control
- VII) Error Handling and Diagnostics

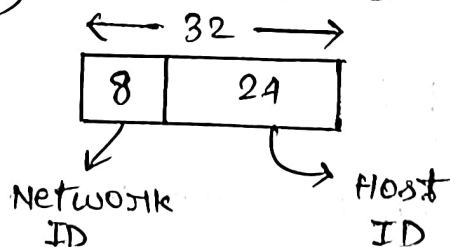
IP Address & Class :

In the Traditional Classful IP Addressing System, there are 5 Class

<u>Class</u>	<u>First Octate Range</u>	<u>Purpose</u>
A	1 - 126	Very large Network
B	128 - 191	Mid Sized Network
C	192 - 223	Small Network
D	224 - 239	Multicasting
E	240 - 255	Experimental / Research

Class A

In IPv4, address has 32 bits.



The first bit is always 0.

- first Octet Range : 1 to 126

⇒ Why not 0 and 127?

- Because 0.x.x.x is reserved for special use.
- 127.x.x.x is a loopback address.

- Number of Network :

Since the first bit is fixed as 0, there are 7 bits available for Network ID,

$$2^7 - 2 = 126$$

So, there are total 126 Class A Networks

- Number of Hosts per Network :

Host Bits = 24

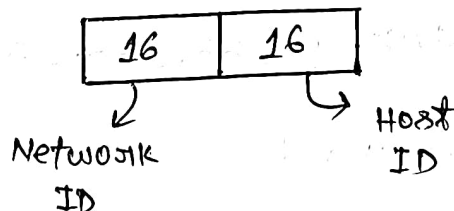
usable Hosts : $(2^{24} - 2) = 16,777,214$

[-2 because one address is the network address and one is the broadcast address]

- Default Subnet Mask : 255.0.0.0

Class B

An IPv4 Address has 32 bits



The first two bits are always 10.

- Range : 128.0.0.0 to 191.255.255.255

- Default Subnet Mask : 255.255.0.0

- Number of Networks:

first two bits are fixed as 10.
 Remaining bits in the first two octets available for Network Number = 14

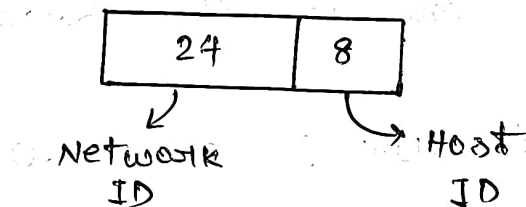
So, Number of Networks, $2^{14} = 16384$

- Number of Hosts per Network:

for class B, Host bits are, 16
 usable Hosts $= (2^{16} - 2) = 65534$

Class C

IPv4 has 32 bit, for class C $\rightarrow$



The first 3 bits are always 110.

- Range: 192.0.0.0 to 223.255.255.255
- Default Subnet Mask: 255.255.255.0
- Number of Networks:

first 3 bits are fixed as 110
 Remaining bits are available = $(24 - 3)$
 $= 21$

So, No of Networks $= 2^{21} = 2,097,152$

- Number of Hosts per Network :
for Class C, Host bits are, 8
usable hosts $(2^8 - 2) = 254$.

Class D

- ⇒ used for Multicasting
- ⇒ First 4 bits are always 1110
- Range : 224.0.0.0 to 239.255.255.255
- Doesn't have a default subnet mask because, it doesn't identify networks and hosts it only identifies multicast Group.

Class E

- ⇒ used for research and Military purpose
- ⇒ first 4 bits are always 1111
- Range : 240.0.0.0 to 255.255.255.255
- Doesn't have any default subnet mask

Disadvantage of Classful Addressing

I) Wastage of IP Address : This is the biggest Problem of classful IP Addressing. The fixed sizes of classes A, B and C often do not match an organization's needs. for example,

If a company needs 1000 hosts

Class C provides 254 hosts

Class B provides 65,534 hosts.

Now, the Company must take Class B Network, wasting thousands of addresses

II) No Support of VLSM:

Classful system is not flexible, there is no variable customized options and also uses different default mask. This is a inefficient address allocation.

III) Large Routing Table: Router had to store separate route for many Network. This makes Routing tables Large and increases memory usage and processing time

IV) Poor Scalability: As the internet grew, fixed classes become inefficient, managing millions of Network become difficult.

Classless Addressing (CIDR)

Classless addressing is the modern method of assigning IP addresses. It was introduced to solve the problems of Classful addressing. In Classful addressing, standard network size is strictly 8, 16 and 24, there is no option inbetween. But in Classless addressing, we can choose between network size of 1 to 32.

Notation

CIDR uses,

IP Address / prefix length

Ex → 192.168.1.0/24

→ Network Bit = 24

Host Bit = $(32 - 24) = 8$

So, Total client $(2^8 - 2) = 254$

Subnetting

Subnetting is the process of dividing one large network into multiple smaller network called subnets.

Subnetting helps by:

- Reducing Broadcast Domain
- Improving Performance
- Better IP utilization
- Easier Network Management

Example →

Suppose we have,

192.168.1.0/24

Host bit = $(32 - 24) = 8$

Usable Host = $(2^8 - 2) = 254$

Now, Let's say we want a smaller network instead of a big one.

We Borrow 2 bits from Host position.

Originally,

Network bits = 24

Host Bits = 8

Now,

Network bits = 26

Host bits = 6

Number of Subnets = 2^n $n = \text{Borrowed bits}$
 $n = 2$

$$\text{Subnets} = 2^2 = 4$$

Host per subnet,

Remaining Host Bit = 6

$$\text{Hosts per Subnet} = (2^6 - 2) \\ = 62$$

The original Network was,

192.168.1.0/24

Now the 4 subnets are,

192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

192.168.1.192/26

VLSM

VLSM or Variable length Subnet Mask is a subnetting technique that allows us to create subnets of different sizes within same network.

Example :

Suppose we have a Network,

$192.168.1.0/24$

Now, we have to make subnets of,

<u>Department</u>	<u>Host</u>
HR	50
IT	25
Accounts	10
Management	5

S1: Allocate Largest Network first

HR needs 50 Hosts, so we have to find smallest subnet eligible,

<u>Prefix</u>	<u>usable Host</u>
/27	30 (Not eligible as we need 50)
/26	62 ✓

So, we allocate $192.168.1.0/26$

Range : $192.168.1.1 \leftrightarrow 192.168.1.62$

Broadcast : $192.168.1.63$

S2: It needs 25 Hosts,

We assign /27 as it has 30 Hosts

We allocate $192.168.1.64/27$

Range : $192.168.1.65 - 192.168.1.94$

Broadcast : $192.168.1.95$

Similarly for Accounts, 10 Hosts,

We assign /28 with 14 Host.

Allocate : 192.168.1.96 /28

Range : 192.168.1.97 $\leftrightarrow$ 192.168.1.110

Broadcast : 192.168.1.111

for Management, 5 Hosts,

We Assign /29 with 6 Hosts.

Allocate : 192.168.1.112 /29

Range : 192.168.1.113 - 192.168.1.118

Broadcast : 192.168.1.119

Supernetting

Supernetting is the process of **combining multiple smaller networks into one larger network**.

It is the opposite of **subnetting**.

Why Do We Need Supernetting?

Suppose an organization has four Class C networks:

```
192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
```

Without supernetting, routers must maintain **4 separate routing entries**.

With supernetting, these can be combined into:

```
192.168.0.0/22
```

Now the router needs only **1 routing entry**.

What Does Supernetting Do?

Before Supernetting

Network

```
192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
```

Routing table:

```
192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24
```

After Supernetting

Combined network:

```
192.168.0.0/22
```

Routing table:

192.168.0.0/22

Only one entry is needed.

Conditions for Supernetting

The networks must:

1. Be Contiguous (consecutive)

Valid:

192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Not valid:

192.168.0.0/24
192.168.2.0/24

because 192.168.1.0 is missing.

2. Have the Same Prefix Length

Example:

All are /24

Example of Supernetting

Networks:

192.168.0.0/24
192.168.1.0/24
192.168.2.0/24
192.168.3.0/24

Binary (third octet):

0 = 00000000
1 = 00000001
2 = 00000010
3 = 00000011

Common bits:

000000

So the prefix becomes:

/22

Result:

192.168.0.0/22

Link State Routing

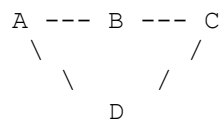
Link State Routing is a routing method where **every router builds a complete map (topology) of the network** and then calculates the shortest path to every destination.

Steps

Step 1: Discover neighbors

Each router finds which routers are directly connected to it.

Example:

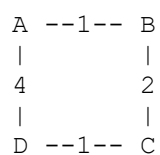


Router A knows it is connected to B and D.

Step 2: Measure link costs

Each router determines the cost of each link (based on bandwidth, delay, etc.).

Example:



Step 3: Create Link State Packet (LSP)

Each router creates a packet containing:

- Router ID
- List of neighbors
- Cost to each neighbor

Example for Router A:

```
Router A:  
B cost = 1  
D cost = 4
```

Step 4: Flood the network

Every router sends its LSP to all routers.

After flooding, **every router has the same network map.**

Step 5: Run Dijkstra's Algorithm

Each router independently runs **Dijkstra's shortest path algorithm** to calculate the best route to every destination.

For Router A:

```
A → B = 1  
A → B → C = 3  
A → B → C → D = 4
```

The router then builds its routing table.

Advantages

- ✓ Fast convergence (adapts quickly to changes)
- ✓ Less chance of routing loops
- ✓ Better for large networks
- ✓ Routers have complete network knowledge

Disadvantages

- ✗ Requires more memory (stores entire topology)
- ✗ More CPU usage (runs Dijkstra)
- ✗ Initial flooding creates overhead

Different types of addressing contained in a packet flowing in the internet

When a packet travels through the Internet, it contains **different types of addresses at different layers of the TCP/IP model**. This is a very common Computer Networks exam question.

1. Physical Address (MAC Address)

Layer: Data Link Layer

- Used for communication within a local network (LAN).
- Every network interface card (NIC) has a unique MAC address.
- Usually 48 bits (6 bytes).

Example:

00:1A:2B:3C:4D:5E

Purpose

When your PC sends data to your router, it uses the router's MAC address as the destination MAC.

Changes?

- ✓ Changes at every hop.

Example:

PC → Router1 → Router2 → Server

The MAC addresses are different on each link.

2. Logical Address (IP Address)

Layer: Network Layer

- Used to identify devices globally on the Internet.
- Routers use IP addresses to forward packets.

Examples:

IPv4: 192.168.1.10

IPv6: 2001:db8::1

Purpose

Tells routers:

- Where the packet came from.
- Where it must go.

Changes?

✗ Normally remains the same from source to destination.

Example:

```
Source IP: 192.168.1.10
Destination IP: 142.250.190.78
```

These stay the same while the packet travels across routers.

3. Port Address

Layer: Transport Layer

- Identifies a specific process or application running on a device.
- Used by TCP and UDP.

Examples:

```
HTTP -> Port 80
HTTPS -> Port 443
FTP -> Port 21
```

Purpose

Many applications may run on the same computer.

Example:

```
Chrome
WhatsApp
Spotify
```

All use the same IP address, so ports identify which application should receive the data.

Example

```
Source Port : 55000
Destination Port : 443
```

4. Specific Address (Application Address)

Layer: Application Layer

Human-readable addresses.

Examples:

- Email address:

abc@gmail.com

- URL:

www.google.com

Purpose

Humans find names easier than IP addresses.

Before communication:

www.google.com
↓ DNS
142.250.190.78

The domain name is converted to an IP address.

Address Type	Layer	Example	Purpose
Physical Address	Data Link	00:1A:2B:3C:4D:5E	Delivery within LAN
Logical Address	Network	192.168.1.10	End-to-end delivery across networks
Port Address	Transport	80, 443, 55000	Identifies application/process
Specific Address	Application	www.google.com, abc@gmail.com	Human-friendly identification

IPv4 (Internet Protocol Version 4)

IPv4 is the most widely used network layer protocol on the Internet. It provides **logical addressing** and **routing** so that data can travel from a source device to a destination device across multiple networks.

What is an IPv4 Address?

An IPv4 address is a **32-bit logical address** assigned to a device on a network.

Example:

192.168.1.10

Internally:

11000000.10101000.00000001.00001010

Since 32 bits = 4 bytes, IPv4 addresses are written as **four decimal numbers separated by dots**.

192.168.1.10
A B C D

Each part (octet) ranges from:

0 to 255

Why Do We Need IPv4?

Suppose you want to send data from Kolkata to Chennai.

Routers need to know:

- Who sent the packet?
- Where the packet should go?

IPv4 provides:

- Source IP address
- Destination IP address

Example:

Source IP : 192.168.1.10
Destination IP : 142.250.190.78

Routers use the destination IP to forward the packet.

Structure of IPv4 Datagram

An IPv4 packet consists of:

```
+-----+
| Header |
+-----+
| Data/Payload |
+-----+
```

The header contains routing information.

IPv4 Header Format

Version	Length (IHL)	Type of Service (TOS)	Total Length			
Identification			Flag	Fragment Offset		
Time To Live (TTL)		Protocol	Header Checksum			
Source IP Address						
Destination IP Address						
Options						

Fields of IPV4

Version

Tells which IP version is being used.

Version = 4

means IPv4.

Source Address

The sender's IP address.

192.168.1.10

This tells the receiver where the packet came from.

Destination Address

The receiver's IP address.

142.250.190.78

Routers use this field to decide where to forward the packet.

TTL (Time To Live)

Prevents packets from looping forever.

Suppose:

TTL = 64

Packet crosses Router 1:

TTL = 63

Router 2:

TTL = 62

and so on.

If TTL reaches 0, the router drops the packet.

Protocol

Tells IPv4 which transport-layer protocol is inside the packet.

Examples:

1 = ICMP

6 = TCP

17 = UDP

If Protocol = 6, IPv4 knows the payload belongs to TCP.

Total Length

Size of the whole packet.

Example:

Header = 20 bytes

Data = 1000 bytes

Then:

Total Length = 1020 bytes

Header Checksum

Used to detect errors in the header.

If a bit gets corrupted during transmission, the checksum helps routers detect it.

Identification, Flags, Fragment Offset

These three fields work together for **fragmentation**.

Imagine a packet is 3000 bytes but a network can only carry 1500-byte packets.

The router breaks it into fragments:

Fragment 1
Fragment 2
Fragment 3

The destination uses:

- Identification → identifies fragments of the same packet
- Flags → tells whether more fragments exist
- Fragment Offset → tells the position of each fragment

so it can rebuild the original packet.

IPv6 (Internet Protocol Version 6)

IPv6 is the successor to IPv4. It was developed because the Internet was running out of IPv4 addresses.

While IPv4 uses **32-bit addresses**, IPv6 uses **128-bit addresses**, providing an enormous number of unique addresses.

Why Was IPv6 Needed?

IPv4 provides:

`2^32 = 4.3 billion addresses`

This seemed huge in the 1980s, but today we have:

- Smartphones
- Laptops
- Smart TVs
- IoT devices
- Servers
- Cameras

IPv4 addresses became scarce.

IPv6 solves this problem.

IPv6 Address Size

IPv6 uses:

`128 bits`

Number of addresses:

`2^128`
`≈ 340 undecillion`
`≈ 3.4 × 10^38`

That's enough for trillions of devices per person on Earth.

IPv6 Address Format

IPv4:

192.168.1.1

IPv6:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv6 addresses are written as:

8 groups
Each group = 16 bits
Hexadecimal digits
Separated by :

Example:

2001:0db8:85a3:0000:0000:8a2e:0370:7334

IPv6 Address Simplification

IPv6 addresses are long, so shortening rules are used.

Rule 1: Remove leading zeros

2001:0db8:0000:0000:0000:ff00:0042:8329

becomes

2001:db8:0:0:0:ff00:42:8329

Rule 2: Consecutive zeros can be replaced by ::

2001:db8:0:0:0:ff00:42:8329

becomes

2001:db8::ff00:42:8329

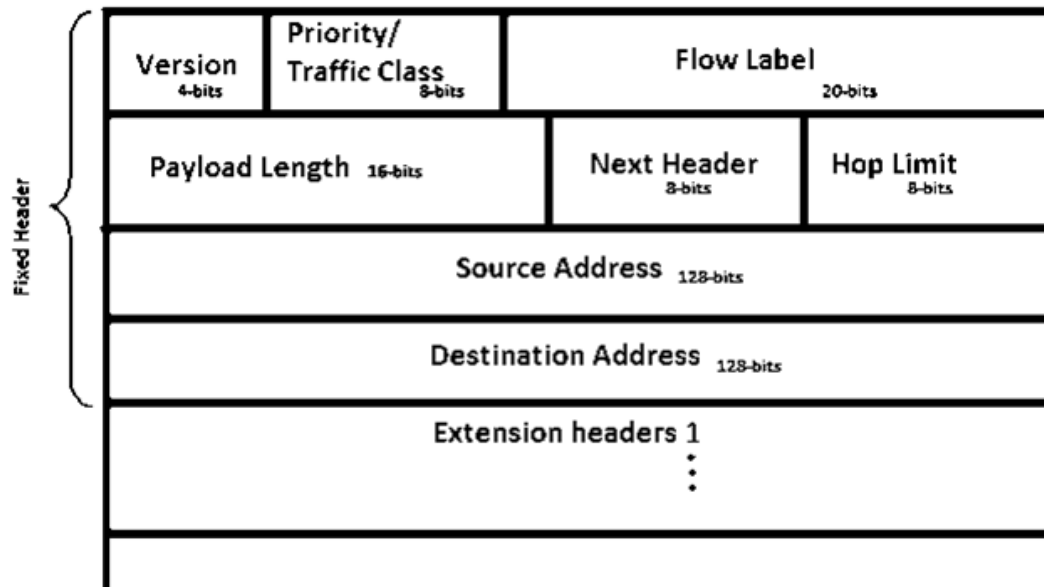
Important:

:: can appear only once in an address.

IPv6 Header Format

One of the biggest improvements over IPv4 is a simpler header.

IPv6 Header:



Fixed size:

40 bytes

Unlike IPv4, which has variable header size.

IPv6 Header Fields

1. Version

Indicates IP version.

Version = 6

2. Traffic Class

Similar to IPv4 ToS field.

Used for:

- Priority
- Quality of Service (QoS)
- Traffic management

Example:

```
Voice call > Email
```

Voice packets can get higher priority.

3. Flow Label

New feature in IPv6.

Used to identify packets belonging to the same flow.

Example:

```
Video streaming  
VoIP call  
Online gaming
```

Routers can handle the entire flow efficiently.

4. Payload Length

Length of actual data.

Does not include header.

5. Next Header

Indicates what comes after IPv6 header.

Examples:

```
6  = TCP  
17 = UDP  
58 = ICMPv6
```

Equivalent to IPv4's Protocol field.

6. Hop Limit

Same purpose as IPv4 TTL.

Example:

```
Hop Limit = 64
```

Each router decreases it by 1.

When it reaches:

0

Packet is discarded.

7. Source Address

128-bit sender address.

Example:

2001:db8::1

8. Destination Address

128-bit receiver address.

Example:

2001:db8::5

Types of IPv6 Addresses

1. Unicast

One sender → One receiver

Example:

PC ----> Server

Most communication uses unicast.

2. Multicast

One sender → Multiple receivers

Example:

Teacher
↓
Students A, B, C

Used for streaming and group communication.

3. Anycast

One sender → Nearest receiver

Example:

User
↓
Nearest DNS Server

Very useful for large services.

No Broadcast in IPv6

IPv4 has broadcast:

255.255.255.255

IPv6 removes broadcast completely.

Uses:

Multicast

instead.

This reduces unnecessary network traffic.

IPv4 vs. IPv6

Feature	IPv4	IPv6
Address Length	32 bits	128 bits
Address Format	Decimal (e.g., 192.168.1.1)	Hexadecimal (e.g., 2001:db8::1)
Address Space	$2^{32} \approx 4.3$ billion	$2^{128} \approx 3.4 \times 10^{38}$
Header Size	Variable (20–60 bytes)	Fixed (40 bytes)
Header Complexity	More complex	Simpler
Checksum Field	Present	Removed
Fragmentation	Routers and hosts can fragment	Only source host fragments
Broadcast	Supported	Not supported
Multicast	Supported	Supported
Anycast	Not available	Available
Security	IPsec optional	IPsec support built-in
Configuration	Manual or DHCP	SLAAC and DHCPv6
NAT	Commonly used	Generally not required
QoS Support	Limited	Better (Traffic Class & Flow Label)
TTL Field	TTL	Hop Limit
Protocol Field	Protocol	Next Header
Address Resolution	ARP	Neighbor Discovery Protocol (NDP)

ICMP (Internet Control Message Protocol)

ICMP is a **Network Layer protocol** used by computers, routers, and other network devices to **report errors and exchange control information**.

It is often called the "**messenger**" of IP because it helps IP communicate problems and network status.

Why Do We Need ICMP?

IP is a **best-effort protocol**.

When something goes wrong, IP itself cannot tell the sender:

- Packet couldn't reach destination
- Router is unreachable
- TTL expired
- Network is congested

ICMP provides these notifications.

Example

Suppose:

Your PC → Router A → Router B → Server

If Router B cannot reach the server:

```
Router B
  ↓
ICMP Error Message
  ↓
Your PC
```

The sender gets informed about the problem.

Position in TCP/IP Model

```
Application Layer
Transport Layer (TCP/UDP)
Network Layer (IP, ICMP)
Data Link Layer
Physical Layer
```

ICMP operates alongside IP at the Network Layer.

How ICMP Works

Suppose you send an IP packet.

```
PC
↓
IP Packet
↓
Router
```

If an error occurs:

```
Router
↓
Creates ICMP Message
↓
Sends back to Source
```

ICMP Message Format

An ICMP message contains:

```
+-----+
| Type   |
+-----+
| Code   |
+-----+
| Checksum |
+-----+
| Data   |
+-----+
```

1. Type Field

Specifies the kind of message.

Examples:

Type	Meaning
0	Echo Reply
3	Destination Unreachable
5	Redirect
8	Echo Request
11	Time Exceeded

2. Code Field

Provides additional details.

Example:

Type 3 (Destination Unreachable)

Codes:

0 → Network unreachable
1 → Host unreachable
3 → Port unreachable

3. Checksum

Used for error detection.

4. Data

Contains information related to the original packet.

Advantages of ICMP over basic IPv4 functionality

1. Error Reporting

- IPv4 does not notify the sender when a problem occurs.
- ICMP sends messages such as:
 - Destination Unreachable
 - Time Exceeded
 - Parameter Problem

2. Network Diagnostics

- ICMP enables tools like:
 - ping
 - traceroute
- IPv4 alone cannot perform these diagnostics.

3. Route Information

- ICMP Redirect messages can inform hosts of a better route.

4. Network Status Feedback

- ICMP provides feedback about packet delivery and network conditions.
- IPv4 only attempts delivery without reporting status.

5. Troubleshooting Support

- Network administrators can identify faults and connectivity issues using ICMP messages.

What is a Gateway?

A **gateway** is a network device that acts as an **entry and exit point between two different networks**. It allows devices on one network to communicate with devices on another network.

Think of it as a **translator and traffic controller** between networks.

Simple Example

Suppose your PC is connected to your home Wi-Fi:

PC → Router (Gateway) → Internet

Your PC cannot directly reach websites on the Internet. It sends packets to the **default gateway** (usually your router), which forwards them to the destination.

Real-Life Analogy

Imagine your neighborhood has only local roads.

To travel to another city, you must first reach the **highway entrance**.

House → Highway Entrance → Other City

The highway entrance is like a **gateway**.

Functions of a Gateway

1. **Connects different networks**
 - LAN ↔ Internet
 - Company Network ↔ Internet
2. **Routes packets**
 - Decides where packets should go next.
3. **Protocol conversion**
 - Can translate between different communication protocols.
4. **Acts as a default route**
 - If a device doesn't know where a destination is, it sends the packet to the gateway.

Default Gateway

Every device usually has:

- IP Address
- Subnet Mask
- Default Gateway

Example:

IP Address : 192.168.1.10
Subnet Mask : 255.255.255.0
Default Gateway : 192.168.1.1

Here, 192.168.1.1 is typically the router.

Gateway vs Router

Feature	Router	Gateway
Purpose	Routes packets between networks	Connects different networks and may translate protocols
OSI Layer	Mainly Network Layer (Layer 3)	Can operate at any layer
Protocol Conversion	Usually No	Yes
Main Function	Forward packets based on IP addresses	Forward + translate communication between different systems
Network Types	Similar networks using same protocol	Different networks/protocols
Complexity	Simpler	More complex
Example	Home Wi-Fi Router	Email gateway, VoIP gateway, Internet gateway

Intranet

An **Intranet** is a **private network that uses Internet technologies (TCP/IP, HTTP, web browsers, etc.) but is accessible only to authorized users within an organization.**

Think of it as a **private Internet for a company, college, bank, or organization.**

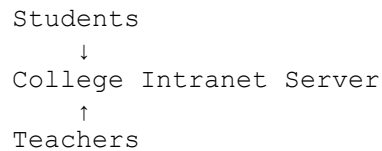
Example

Suppose your college has a website where students can:

- Check attendance
- View marks
- Download notes
- Access notices

and this website is only accessible to students and staff.

That is an **Intranet**.



Characteristics of Intranet

1. Private Network

Only authorized users can access it.

2. Uses Internet Technologies

Uses:

- TCP/IP
- HTTP/HTTPS
- Web browsers
- Email

Just like the Internet.

3. Secure

Protected by:

- Login credentials
- Firewalls
- Access control

4. Internal Communication

Used for sharing information within an organization.

Uses of Intranet

- Employee communication
- Attendance management
- Internal emails
- Sharing documents
- Company announcements
- Project management

MAC Address vs IP Address

Both are used to identify devices in a network, but they serve different purposes.

Feature	MAC Address	IP Address
Full Form	Media Access Control Address	Internet Protocol Address
Layer	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Purpose	Identifies a network interface (NIC)	Identifies a device on a network
Assigned By	Manufacturer	Network administrator, ISP, DHCP
Length	48 bits (usually)	IPv4: 32 bits, IPv6: 128 bits
Format	Hexadecimal	Decimal (IPv4) / Hexadecimal (IPv6)
Changes?	Usually permanent	Can change
Scope	Local Network (LAN)	Local + Internet

Connection-oriented and Connectionless service

Connection-oriented service establishes a logical connection between sender and receiver before data transmission and provides reliable, ordered delivery with error control. TCP is an example of a connection-oriented protocol.

Connectionless service sends packets independently without establishing a connection and does not guarantee delivery, ordering, or reliability. UDP is an example of a connectionless protocol.

Feature	Connection-Oriented Service	Connectionless Service
Definition	Connection is established before data transfer	No connection is established before data transfer
Connection Setup	Required	Not Required
Connection Termination	Required	Not Required
Reliability	Reliable	Unreliable
Acknowledgement (ACK)	Used	Not Used
Error Recovery	Supported through retransmission	Not Supported
Packet Delivery Guarantee	Guaranteed	Not Guaranteed

Packet Ordering	Packets arrive in order	Order not guaranteed
Flow Control	Available	Not Available
Congestion Control	Available	Not Available
Overhead	High	Low
Speed	Slower	Faster
Resource Usage	More memory and processing	Less memory and processing
Suitable For	Applications requiring reliability	Real-time applications
Example Protocol	TCP	UDP
Real-Life Analogy	Phone call 📞	Postal letter 📬

Static and Dynamic Routing

Static Routing

In static routing, the administrator manually specifies routes.

Example:

```
Destination: 192.168.2.0/24
Next Hop: 10.0.0.2
```

If the network changes:

```
Link Failure
↓
Administrator must update routes manually
```

Advantages

- Simple for small networks
- More secure
- No routing protocol overhead
- Predictable routing behavior

Disadvantages

- Difficult to manage in large networks
- No automatic recovery from failures
- Time-consuming maintenance

Dynamic Routing

Routers automatically exchange routing information using routing protocols.

Example:

Router A ↔ Router B ↔ Router C

If a link fails:

```
Router A
  X
Router B
  |
Router C
```

Routers automatically calculate a new path.

Advantages

- Automatic route discovery
- Adapts to network changes
- Suitable for large networks
- Better fault tolerance

Disadvantages

- Consumes CPU, memory, and bandwidth
- More complex
- Routing updates create overhead

Why is Dynamic Routing Preferred Over Static Routing?

1. **Automatic Route Updates**
Routers automatically learn and update routes when the network changes.
2. **Handles Failures Automatically**
If a link fails, routers find an alternative path without manual intervention.
3. **Suitable for Large Networks**
Managing routes manually becomes difficult in large networks; dynamic routing scales easily.
4. **Less Administrative Effort**
No need to manually add, delete, or modify routes frequently.
5. **Supports Network Growth**
New routers and networks are learned automatically.
6. **Better Reliability**
Network remains operational even when some links or routers fail.

Static Routing Vs. Dynamic Routing

Feature	Static Routing	Dynamic Routing
Definition	Routes are manually configured by the network administrator	Routes are learned automatically using routing protocols
Route Updates	Manual	Automatic
Administrative Effort	High	Low
Adaptation to Network Changes	Does not adapt automatically	Adapts automatically
Routing Overhead	Very low	Higher (routing updates consume bandwidth and CPU)
Scalability	Suitable for small networks	Suitable for medium and large networks
Fault Tolerance	Poor	Good
Routing Table Maintenance	Manual	Automatic
CPU & Memory Usage	Less	More
Security	More secure (no routing advertisements)	Less secure unless protected
Convergence	No convergence process	Converges automatically after topology changes
Network Traffic	No routing update traffic	Generates routing update traffic
Configuration Complexity	Simple for small networks	More complex to configure initially
Examples	Static routes configured by admin	OSPF, RIP, EIGRP, IS-IS

Link State Routing

Link State Routing is a dynamic routing technique in which **every router builds a complete map (topology) of the network and then calculates the shortest path to all destinations using Dijkstra's Algorithm.**

Unlike Distance Vector Routing, routers do **not** simply tell neighbors the distance to destinations. Instead, they share information about their directly connected links.

Concept

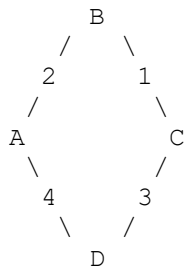
Every router asks:

Who are my neighbors?
What is the cost of reaching them?

Then it shares this information with all other routers.

As a result, every router eventually has the same network map.

Example Network



Numbers represent link costs.

How Link State Routing Works

Step 1: Discover Neighbors

Router A discovers:

```
Neighbor B (cost 2)
Neighbor D (cost 4)
```

Step 2: Measure Link Costs

Each router determines the cost of its links.

Cost may depend on:

- Bandwidth
- Delay
- Congestion
- Administrator settings

Example:

```
A → B = 2
A → D = 4
```

Step 3: Create Link State Packet (LSP)

Each router prepares a Link State Packet containing:

```
Router ID
List of neighbors
Cost to each neighbor
Sequence Number
Age
```

Example for Router A:

```
Router A:
B cost 2
D cost 4
```

Step 4: Flooding

The LSP is sent to all routers in the network.

```
A → B
A → D
```

Then B and D forward it further.

Eventually every router receives A's LSP.

The same happens for all routers.

Step 5: Build Link State Database (LSDB)

After flooding:

Each router stores all LSPs.

Example:

```
Router A knows:
A-B cost 2
A-D cost 4
B-C cost 1
C-D cost 3
```

Now A has the complete network map.

Step 6: Run Dijkstra's Algorithm

Each router independently calculates the shortest path tree.

For Router A:

Possible paths to C:

```
A → B → C = 2 + 1 = 3
A → D → C = 4 + 3 = 7
```

Shortest path:

```
A → B → C
Cost = 3
```

Distance Vector Routing

Distance Vector Routing is a dynamic routing technique in which each router maintains a table containing:

1. **Destination Network**
2. **Distance (Cost/Metric)**
3. **Direction (Next Hop Router)**

Routers periodically exchange their routing tables with their immediate neighbors and update routes using the **Bellman-Ford Algorithm**.

Why is it called "Distance Vector"?

Distance

The cost to reach a destination.

Examples:

Hop Count
Delay
Bandwidth Cost

Vector

The direction (next hop) to reach the destination.

Example:

Destination: Network D
Distance: 3 hops
Vector: Router B

Meaning:

"To reach D, go through Router B and it will take 3 hops."

Example Network

```
A ----- B ----- C
          |
          |
          D
```

Initially:

Router B knows

A = 1 hop
C = 1 hop
D = 1 hop

Router A knows

B = 1 hop

A doesn't know about C and D yet.

How Distance Vector Routing Works

Step 1: Create Routing Table

Router A:

Destination	Distance	Next Hop
A	0	-
B	1	B

Step 2: Exchange Routing Tables

Router B sends its routing table to A.

B says:

I can reach:
A = 1 hop
C = 1 hop
D = 1 hop

Step 3: Update Routing Table

Router A calculates:

A → B → C
Cost = 1 + 1 = 2 hops
A → B → D
Cost = 1 + 1 = 2 hops

New table:

Destination	Distance	Next Hop
A	0	-
B	1	B
C	2	B

D	2	B
---	---	---

Step 4: Periodic Updates

Routers continue sharing routing tables periodically.

Eventually every router learns routes to every destination.

RIP and BGP

Both **RIP** and **BGP** are routing protocols, but they are used in very different situations.

1. RIP (Routing Information Protocol)

RIP is a **Distance Vector Routing Protocol**.

It uses **hop count** as the metric to determine the best path.

How RIP Works

Routers periodically exchange their routing tables with neighboring routers.

Example:

A ---- B ---- C

Router A learns about C through B.

Metric Used

RIP measures distance using:

Hop Count

Example:

A → B → C

Distance:

2 hops

The route with the fewest hops is chosen.

Maximum Hop Count

15 hops

If hop count becomes:

16 hops

the destination is considered unreachable.

Characteristics of RIP

Feature	RIP
Type	Distance Vector
Algorithm	Bellman-Ford
Metric	Hop Count
Max Hops	15
Updates	Every 30 seconds
Convergence	Slow
Network Size	Small Networks

Advantages

- Simple
- Easy to configure
- Low resource requirements

Disadvantages

- ✚ Slow convergence
- ✚ Routing loops possible
- ✚ Maximum 15 hops
- ✚ Not suitable for large networks

2. BGP (Border Gateway Protocol)

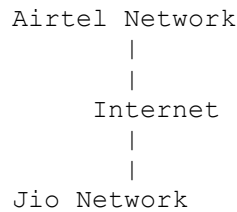
BGP is the routing protocol that powers the Internet.

It is a **Path Vector Routing Protocol**.

Used between different organizations or Internet Service Providers (ISPs).

Example

Suppose:



How does Airtel know how to reach Jio?

Using BGP.

Autonomous System (AS)

An **Autonomous System (AS)** is a large network managed by a single organization.

Examples:

- ISP networks
- Google network
- Microsoft network
- University networks

Each AS has a unique AS Number (ASN).

How BGP Works

Instead of just saying:

Distance = 5 hops

BGP says:

Path:
AS100 → AS200 → AS300

It advertises the complete AS path.

Example

AS100 → AS200 → AS300 → Destination

BGP can see the entire path and avoid routing loops.

Characteristics of BGP

Feature	BGP
Type	Path Vector
Scope	Internet
Metric	AS Path, Policies, Attributes

Convergence	Slower but stable
Scalability	Extremely High
Usage	Between Autonomous Systems

Advantages

- Internet-scale routing
- Loop prevention
- Highly scalable
- Policy-based routing

Disadvantages

- ✚ Complex configuration
- ✚ Slower convergence
- ✚ Higher administrative overhead

RIP vs BGP

Feature	RIP	BGP
Full Form	Routing Information Protocol	Border Gateway Protocol
Routing Type	Distance Vector	Path Vector
Algorithm	Bellman-Ford	Path Vector
Metric	Hop Count	AS Path & Policies
Maximum Hops	15	No such limit
Scalability	Small Networks	Internet-scale
Convergence	Faster than BGP but less stable	Slower but more stable
Use Case	Inside small organizations	Between ISPs and large networks
Complexity	Simple	Complex

EGP (Exterior Gateway Protocol)

EGP stands for Exterior Gateway Protocol.

It is a routing protocol used to exchange routing information **between different Autonomous Systems (ASes).**

Why Do We Need EGP?

Inside an Autonomous System, routers use protocols like:

- RIP

- OSPF

These are called **Interior Gateway Protocols (IGPs)**.

But when different Autonomous Systems need to communicate:

```
Airtel AS
  ↓
Google AS
  ↓
Jio AS
```

an Exterior Gateway Protocol is needed.

Interior vs Exterior Routing

Interior Routing

Within the same AS:

```
Router A
  ↓
Router B
  ↓
Router C
```

Uses:

- RIP
- OSPF
- IS-IS

Exterior Routing

Between different ASes:

```
AS100
  ↓
AS200
  ↓
AS300
```

Uses EGP.

Historical EGP

Originally there was an actual protocol named:

```
EGP
(Exterior Gateway Protocol)
```

developed in the early Internet.

However, it had limitations:

- Poor scalability
- Could not handle today's Internet

It is now obsolete.

Modern EGP = BGP

Today when people say:

EGP

they usually mean:

BGP
(Border Gateway Protocol)

because BGP is the only widely used Exterior Gateway Protocol on the Internet.

ARP & RARP

ARP

Address Resolution Protocol (ARP) is a protocol used to obtain the MAC address corresponding to a known IPv4 address on a local network.

RARP

Reverse Address Resolution Protocol (RARP) is a protocol used to obtain an IP address corresponding to a known MAC address. It has largely been replaced by DHCP.

Feature	ARP	RARP
Full Form	Address Resolution Protocol	Reverse Address Resolution Protocol
Purpose	Finds MAC address from IP address	Finds IP address from MAC address
Mapping	IP → MAC	MAC → IP
Used By	Hosts sending packets on a LAN	Diskless workstations (historically)
Request	"I know the IP, what's the MAC?"	"I know my MAC, what's my IP?"
Usage Today	Widely used	Mostly obsolete
Replacement	Still used	Replaced by BOOTP and DHCP

NAT (Network Address Translation)

NAT (Network Address Translation) is a technique used by a router to **translate private IP addresses into public IP addresses** and vice versa.

It allows multiple devices in a private network to share a single public IP address when accessing the Internet.

Why is NAT Needed?

IPv4 addresses are limited.

Suppose your home network has:

```
PC       : 192.168.1.2
Laptop   : 192.168.1.3
Phone    : 192.168.1.4
```

These are **private IP addresses** and cannot be used directly on the Internet.

The router has:

```
Public IP = 49.37.XX.XX
```

NAT translates the private IPs into the public IP before sending packets to the Internet.

How NAT Works

```
PC (192.168.1.2)
  ↓
Router (NAT)
  ↓
Internet
```

The router maintains a NAT table:

Private IP	Public IP
192.168.1.2	49.37.XX.XX
192.168.1.3	49.37.XX.XX
192.168.1.4	49.37.XX.XX

When replies come back, the router uses the table to forward them to the correct device.

OSPF (Open Shortest Path First)

OSPF (Open Shortest Path First) is a **Link-State Routing Protocol** used for **dynamic routing within an Autonomous System (AS)**.

It finds the **shortest path** between source and destination using **Dijkstra's Shortest Path Algorithm (SPF Algorithm)**.

Features of OSPF

- Dynamic routing protocol
- Link-state routing protocol
- Uses **cost** as routing metric
- Fast convergence
- Supports large networks
- Supports VLSM and CIDR
- Uses Dijkstra's algorithm

How OSPF Works

1. Discover Neighbors

Routers exchange **Hello packets** to discover neighboring routers.

2. Exchange Link-State Information

Each router creates a **Link State Advertisement (LSA)** containing information about its links and costs.

3. Flood LSAs

LSAs are flooded throughout the OSPF area.

4. Build Link-State Database (LSDB)

Every router obtains the same network topology map.

5. Calculate Shortest Path

Each router runs **Dijkstra's Algorithm** and creates its routing table.

ICMP (Internet Control Message Protocol)

ICMP is a Network Layer protocol used for **error reporting, diagnostics, and network status messages** in IP networks.

It works alongside IPv4 and IPv6 and helps devices determine whether packets are being delivered correctly.

Why is ICMP Needed?

IP is a **connectionless protocol** and does not report errors itself.

For example:

PC — Router — Server

If the server is unreachable, IP alone cannot inform the sender.

ICMP sends an error message such as:

Destination Unreachable

back to the sender.

Main Functions of ICMP

1. Error Reporting

Examples:

- Destination Unreachable
- Time Exceeded
- Parameter Problem

2. Network Diagnostics

Used by:

- Ping
- Traceroute

3. Network Information

Provides feedback about network conditions.

Transport Layer

Congestion

Congestion is a condition in a network where the **number of packets arriving exceeds the network's capacity to handle them**, causing delays, packet loss, and reduced performance.

Definition

Congestion occurs when too much traffic is present in the network and routers/switches cannot process or forward packets fast enough.

Example

Imagine a router can handle:

1000 packets/second

but receives:

2000 packets/second

The extra packets start filling the buffer (queue).

When the buffer becomes full:

Packets are dropped

This is congestion.

Effects of Congestion

- Increased delay
- Packet loss
- Reduced throughput
- Retransmissions
- Poor network performance

Causes of Congestion

1. Too many users sending data simultaneously.
2. Insufficient bandwidth.
3. Slow routers or network devices.
4. Burst traffic (sudden increase in packets).

Congestion Control Techniques

- Traffic shaping
- Admission control
- Leaky Bucket Algorithm
- Token Bucket Algorithm
- TCP Congestion Control

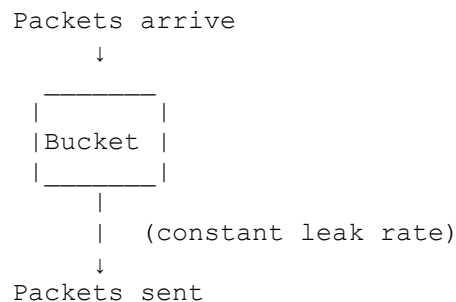
Leaky Bucket Algorithm

The **Leaky Bucket Algorithm** is a **traffic shaping and congestion control** technique used to control the rate at which packets are sent into a network.

The idea comes from a bucket with a small hole at the bottom.

Basic Idea

Imagine a bucket:



- Packets enter the bucket at any rate.
- Packets leave the bucket at a **fixed, constant rate**.
- If the bucket becomes full, newly arriving packets are discarded.

Working

Suppose:

- Bucket size = 10 packets
- Output rate = 2 packets/second

Case 1: Normal Traffic

Arrival Rate = 2 packets/sec

Packets leave as fast as they arrive.

No congestion.

Case 2: Burst Traffic

Arrival Rate = 8 packets/sec

Packets are stored in the bucket.

Output remains:

2 packets/sec

The burst is smoothed into a steady flow.

Case 3: Bucket Full

Bucket Capacity Reached

New packets arriving are dropped.

This prevents network overload.

Advantages

- Controls congestion.
- Smooths bursty traffic.
- Produces a constant output rate.
- Simple to implement.

Disadvantages

- Packets may be dropped when the bucket is full.
- Does not allow large bursts, even if the network could handle them.
- Can waste available bandwidth.

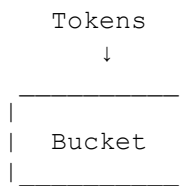
Token Bucket Algorithm

The **Token Bucket Algorithm** is a **traffic shaping and congestion control** technique that controls data transmission using **tokens**.

Unlike the Leaky Bucket Algorithm, it **allows bursty traffic** while still controlling the average transmission rate.

Basic Idea

Imagine a bucket that stores tokens:



- Tokens are added to the bucket at a fixed rate.
- To send a packet, a device must remove (consume) tokens.
- If enough tokens are available, the packet is sent immediately.
- If no tokens are available, the packet must wait or be dropped.

Working

Suppose:

- Token generation rate = 5 tokens/sec
- Bucket capacity = 20 tokens
- 1 token allows transmission of 1 packet

Normal Traffic

5 packets/sec

Tokens are consumed as they are generated.

Bursty Traffic

If the bucket has accumulated:

20 tokens

a device can suddenly send:

20 packets

at once.

This is why Token Bucket supports bursts.

Example

Token Rate = 10 tokens/sec
Bucket Size = 50 tokens

If the network is idle for 5 seconds:

50 tokens collected

Now a burst of 50 packets can be transmitted immediately.

Advantages

- Allows bursty traffic.
- Better bandwidth utilization.
- Flexible and efficient.
- Widely used in modern networks.

Disadvantages

- Slightly more complex than Leaky Bucket.
- Packets may still be delayed or dropped if tokens are exhausted.

Admission Control

Admission Control is a congestion control technique that decides **whether a new traffic flow or connection should be allowed into the network** based on the availability of network resources.

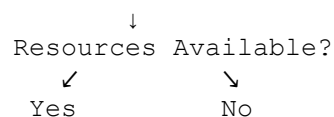
The main goal is to **prevent congestion before it occurs**.

How It Works

When a new connection request arrives:

1. The network checks available resources (bandwidth, buffer space, etc.).
2. If sufficient resources are available → **Accept the connection**.
3. If resources are insufficient → **Reject or delay the connection**.

New Connection Request



↓ ↓
Accept Reject

Example

Suppose a router can handle:

100 Mbps

Current usage:

90 Mbps

A new video stream requires:

20 Mbps

Since:

$90 + 20 = 110$ Mbps

which exceeds capacity, the network rejects the request.

This prevents congestion.

Advantages

- Prevents network congestion.
- Maintains QoS (Quality of Service).
- Ensures existing connections are not affected.
- Improves network reliability.

TCP Congestion Control

TCP Congestion Control is a mechanism used by TCP to prevent network congestion by controlling the amount of data sent into the network.

TCP adjusts its sending rate based on network conditions to avoid overloading routers and links.

Why is it Needed?

When too many packets are sent:

- Router buffers fill up.

- Packets are dropped.
- Delays increase.
- Network performance decreases.

TCP detects congestion and reduces its transmission rate.

Main Techniques in TCP Congestion Control

1. Slow Start

At the beginning of a connection, TCP does not know the network capacity.

- Starts with a small **Congestion Window (cwnd)**.
- Doubles the window every RTT (Round Trip Time).

Example:

1 → 2 → 4 → 8 → 16 ...

This quickly discovers available bandwidth.

2. Congestion Avoidance

When cwnd reaches a threshold (**ssthresh**):

TCP grows more slowly.

16 → 17 → 18 → 19 ...

Instead of doubling, it increases linearly.

This avoids sudden congestion.

3. Fast Retransmit

If the sender receives **3 duplicate ACKs**:

TCP assumes a packet was lost.

Instead of waiting for a timeout:

```

Lost Packet
  ↓
Immediately Retransmit

```

This improves performance.

4. Fast Recovery

After Fast Retransmit:

- TCP reduces the congestion window.
- Does not return to the beginning (Slow Start).
- Continues transmission at a reduced rate.

This helps recover faster from packet loss.

TCP & UDP

TCP (Transmission Control Protocol)

TCP is a **connection-oriented Transport Layer protocol** that provides **reliable and error-free communication** between devices.

Before sending data, TCP establishes a connection using the **3-way handshake**.

Features of TCP

- Connection-oriented
- Reliable data delivery
- Uses acknowledgments (ACKs)
- Retransmits lost packets
- Maintains packet order
- Provides flow control
- Provides congestion control

Working of TCP (3-Way Handshake)

```
Client          Server
|  ----  SYN  ---->  |
|  <---  SYN+ACK  --  |
|  ----  ACK  ---->  |
```

After the connection is established, data transfer begins.

Applications of TCP

- Web browsing (HTTP/HTTPS)
- File Transfer (FTP)
- Email (SMTP, POP3, IMAP)
- Remote login (SSH)

UDP (User Datagram Protocol)

UDP is a **connectionless Transport Layer protocol** that sends data without establishing a connection. It is faster than TCP because it does not use acknowledgments, retransmissions, or flow control.

Features of UDP

- Connectionless
- Unreliable delivery
- No acknowledgments
- No retransmission
- No packet ordering guarantee
- Low overhead
- Faster than TCP

Working of UDP

Sender -----> Receiver
(Data sent directly)

Packets are sent immediately without connection setup.

Applications of UDP

- Online gaming
- Video streaming
- VoIP calls
- DNS

Feature	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
Connection Type	Connection-Oriented	Connectionless
Reliability	Reliable	Unreliable
Acknowledgment	Uses ACKs	No ACKs
Error Recovery	Yes	No
Packet Ordering	Maintains order	No guarantee of order
Flow Control	Yes	No
Congestion Control	Yes	No
Speed	Slower	Faster
Header Size	20–60 bytes	8 bytes
Overhead	High	Low
Applications	Web, Email, File Transfer	Streaming, Gaming, DNS, VoIP

QoS (Quality of Service)

Quality of Service (QoS) refers to the ability of a network to provide **better and predictable service** to different types of traffic by managing bandwidth, delay, jitter, and packet loss.

QoS is especially important for real-time applications such as voice calls, video conferencing, and online gaming.

Why QoS is Needed

Different applications have different requirements:

- Email → can tolerate delay.
- File download → needs throughput.
- Voice call → needs low delay and low jitter.
- Video conferencing → needs low delay and low packet loss.

QoS ensures that important traffic gets priority.

QoS Parameters

1. Bandwidth

Amount of data that can be transmitted per second.

Higher bandwidth → More data transfer

2. Delay (Latency)

Time taken for a packet to travel from source to destination.

Lower delay = Better performance

3. Jitter

Variation in packet delay.

Example:

Packet 1 → 20 ms
Packet 2 → 40 ms
Packet 3 → 15 ms

The variation is jitter.

High jitter causes choppy audio/video.

4. Packet Loss

Packets dropped during transmission.

More packet loss → Poor quality

QoS Techniques

- **Traffic Classification** – Identify different types of traffic.
- **Priority Queuing** – Give important traffic higher priority.
- **Traffic Shaping** – Control data transmission rate.
- **Resource Reservation** – Reserve bandwidth for critical applications.

Application Layer

What is Data Privacy?

Data Privacy refers to the protection of data from unauthorized access, disclosure, or misuse. It ensures that only authorized users can view or access sensitive information.

Example

If you send a message to a friend:

"Meet me at 5 PM"

Only your friend should be able to read it. If an attacker reads it, privacy is violated.

Methods to Achieve Data Privacy

- Encryption (AES, DES, RSA)
- Access Control
- Authentication Mechanisms
- Secure Communication Protocols (HTTPS, TLS)

Definition

Data Privacy is the protection of information from unauthorized access, disclosure, modification, or misuse, ensuring that only authorized entities can access the data.

Digital Signature

A **Digital Signature** is a cryptographic technique used to verify the sender's identity and ensure that a message has not been altered.

Basic Process

1. Sender computes a hash of the message.
2. Sender encrypts the hash using their **private key**.
3. This encrypted hash is the **digital signature**.
4. Receiver decrypts the signature using the sender's **public key** and compares it with a newly computed hash.

Message

↓

Hash

↓
Encrypt with Private Key
↓
Digital Signature

How Digital Signatures Provide Authentication

Authentication means verifying the identity of the sender.

How?

Only the sender possesses their private key.

Sender's Private Key
↓
Digital Signature

If the signature is successfully verified using the sender's public key, the receiver knows that the message was signed by that sender.

✅ Sender identity is verified.

How Digital Signatures Provide Integrity

Integrity means the message has not been modified during transmission.

How?

The receiver:

1. Calculates the hash of the received message.
2. Decrypts the digital signature to obtain the original hash.
3. Compares the two hashes.

Hash1 = Calculated Hash
Hash2 = Decrypted Hash

Hash1 = Hash2 ?

If both hashes match:

Message unchanged

If they differ:

Message altered

✓ Integrity is ensured.

How Digital Signatures Provide Non-Repudiation

Non-repudiation means the sender cannot later deny sending the message.

How?

Since the signature was created using the sender's private key:

Only Sender Knows Private Key

the sender cannot claim:

"I never sent this message."

because anyone can verify that the signature was generated using the sender's private key.

✓ Sender cannot deny the action.

DNS (Domain Name System)

DNS (Domain Name System) is a naming service that translates **human-readable domain names** into **IP addresses**.

It is often called the "**phonebook of the Internet**."

Why is DNS Needed?

Humans prefer names like:

www.google.com
www.youtube.com

But computers communicate using IP addresses:

142.250.xxx.xxx

DNS converts the domain name into the corresponding IP address.

How DNS Works

Suppose you enter:

`www.google.com`

Steps:

1. Browser asks the DNS server:

"What is the IP address of `www.google.com`?"
2. DNS server looks up the address.
3. DNS server replies with the IP address.
4. Browser connects to that IP address.

```
www.google.com
  ↓
  DNS
  ↓
142.250.xx.xx
  ↓
Google Server
```

DNS Hierarchy

DNS is organized in a hierarchical structure:

```
Root DNS Server
  ↓
Top Level Domain (TLD)
  ↓
Authoritative DNS Server
```

Example:

```
www.google.com
  ↓
.com (TLD)
  ↓
google
```

Components of DNS

1. Root Server

Topmost DNS server.

2. TLD Server

Handles domains like:

.com
.org
.edu
.in

3. Authoritative DNS Server

Stores the actual IP address of the domain.

Advantages of DNS

- Easy to remember website names.
- Eliminates the need to memorize IP addresses.
- Scalable and distributed.
- Faster access through DNS caching.

Symmetric & Asymmetric Cryptography

1. Symmetric Key Cryptography

In **Symmetric Key Cryptography**, the **same key** is used for both encryption and decryption.

```
Plaintext
  ↓ (Key K)
Encryption
  ↓
Ciphertext
  ↓ (Same Key K)
Decryption
  ↓
Plaintext
```

Example

Alice and Bob share a secret key:

Key = SECRET123

- Alice encrypts using SECRET123.
- Bob decrypts using SECRET123.

Advantages

- Fast
- Efficient for large amounts of data
- Less computational overhead

Disadvantages

- Key distribution is difficult.
- If the key is leaked, security is compromised.

Examples

- Advanced Encryption Standard (AES)
- Data Encryption Standard (DES)
- Triple DES

2. Asymmetric Key Cryptography

In **Asymmetric Key Cryptography**, two different keys are used:

1. **Public Key** (shared with everyone)
2. **Private Key** (kept secret)

Public Key → Encrypt
Private Key → Decrypt

Example

Bob has:

Public Key = Available to everyone
Private Key = Secret

Alice encrypts a message using Bob's public key.

Only Bob can decrypt it using his private key.

```
Alice
↓
Encrypt with Bob's Public Key
↓
Ciphertext
↓
Bob
↓
Decrypt with Private Key
```

Advantages

- Solves the key distribution problem.
- Provides digital signatures and non-repudiation.

Disadvantages

- Slower than symmetric encryption.
- Requires more computation.

Examples

- RSA cryptosystem
- Diffie–Hellman key exchange
- Elliptic Curve Cryptography (ECC)

Symmetric vs Asymmetric Cryptography

Feature	Symmetric	Asymmetric
Number of Keys	One	Two
Encryption Key	Same as decryption key	Different from decryption key
Speed	Faster	Slower
Security	Depends on secret key sharing	More secure key management
Key Distribution	Difficult	Easy
Examples	AES, DES	RSA, ECC

KDC (Key Distribution Center)

KDC (Key Distribution Center) is a trusted third party that securely distributes secret keys to users who want to communicate using **symmetric-key cryptography**.

Its main purpose is to solve the **key distribution problem**.

Why is KDC Needed?

In symmetric encryption:

Alice ↔ Bob

both need the same secret key.

The problem is:

How can Alice and Bob securely obtain the same key?

This is where the KDC helps.

Working of KDC

Suppose Alice wants to communicate with Bob.

Step 1

Alice and KDC share a secret key:

K_A

Bob and KDC share another secret key:

K_B

Step 2

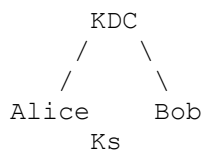
Alice requests a session key for communication with Bob.

Step 3

KDC generates a **Session Key (K_s)**.

Step 4

KDC securely sends K_s to Alice and Bob.



Step 5

Alice and Bob use:

Session Key K_s

to encrypt and decrypt messages.

Key Agreement & Key Distribution

1. Key Distribution (Key Transport) Protocol

A **Key Distribution/Key Transport Protocol** is a method where **one party generates the secret key and securely sends (transports) it to the other party.**

Example

```
Alice generates key K
      ↓
Alice securely sends K
      ↓
Bob receives K
```

Both now use the same key K for encryption and decryption.

Examples

- Using a trusted Key Distribution Center (KDC)
- RSA key transport

Exam Definition

Key Distribution (or Key Transport) is a process in which one entity generates a secret key and securely transfers it to another entity.

2. Key Agreement Protocol

A **Key Agreement Protocol** is a method where **both parties contribute information to create a shared secret key.**

No single party chooses the entire key.

Example

```
Alice contributes value A
Bob contributes value B

      ↓

Shared Secret Key K
```

Neither Alice nor Bob alone determines the final key.

Example

The classic example is the Diffie–Hellman key exchange.

Alice Secret $\rightarrow a$
Bob Secret $\rightarrow b$

After exchange:

Shared Key = $g^{(ab)}$

Both obtain the same key without sending the key itself across the network.

Definition

Key Agreement is a process in which two or more parties jointly establish a secret key, such that no single party can determine the entire key alone.

Key Agreement vs Key Distribution

Feature	Key Distribution (Transport)	Key Agreement
Who creates the key?	One party	Both parties
Key sent over network?	Yes (securely)	No, only exchange values
Security	Depends on secure transport	More secure
Example	RSA Key Transport, KDC	Diffie–Hellman
Contribution	One-sided	Both sides

SMTP (Simple Mail Transfer Protocol)

SMTP (Simple Mail Transfer Protocol) is an **Application Layer protocol** used for **sending and transferring emails** over the Internet.

It is responsible for:

- Sending email from a user to a mail server.
- Transferring email between mail servers.

SMTP only sends mail. To receive/read emails, protocols like **POP3** or **IMAP** are used.

How SMTP Works

Suppose Alice sends an email to Bob:

Alice
↓
Mail Client
↓
SMTP Server
↓
Internet
↓
Bob's Mail Server
↓
Bob

Steps:

1. Alice composes an email.
2. Her mail client sends it to the SMTP server.
3. The SMTP server uses DNS to locate Bob's mail server.
4. The email is transferred to Bob's mail server.
5. Bob retrieves it using POP3 or IMAP.

SMTP Commands

Command	Purpose
HELO/EHLO	Identify the sender
MAIL FROM	Specify sender address
RCPT TO	Specify recipient address
DATA	Send message content
QUIT	End the session

SMTP Ports

Port	Use
25	Mail server to mail server
587	Email submission (most common)
465	Secure SMTP (SMTPS)

Advantages

- Reliable email delivery.
- Simple and widely used.
- Supports communication between different mail systems.

Limitation

- Cannot retrieve emails.
- Original SMTP was not encrypted (modern SMTP usually uses TLS).

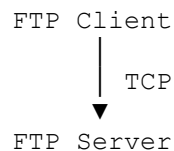
FTP (File Transfer Protocol)

FTP (File Transfer Protocol) is an **Application Layer protocol** used to **transfer files between a client and a server** over a TCP/IP network.

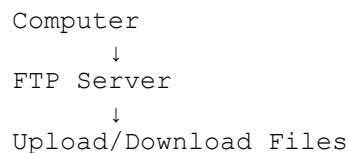
It allows users to:

- Upload files
- Download files
- Delete files
- Rename files
- Manage directories on a remote server

How FTP Works



Example:



FTP Uses Two Connections

1. Control Connection

Used for commands and responses.

- Port **21**
- Carries commands like LOGIN, LIST, GET, PUT

2. Data Connection

Used for actual file transfer.

- Port **20** (traditional FTP)
- Carries files, images, videos, documents, etc.

Client — Port 21 —> Commands
Client — Port 20 —> File Transfer

FTP Working

1. Client connects to the FTP server.
2. User provides username and password.
3. Commands are sent through the control connection.
4. Files are transferred through the data connection.
5. Connection is closed after completion.

Common FTP Commands

Command	Purpose
USER	Enter username
PASS	Enter password
LIST	Show files/directories
GET	Download a file
PUT	Upload a file
DELETE	Remove a file
QUIT	Disconnect

Advantages

- Fast file transfer.
- Supports large files.
- Allows both upload and download.
- Easy remote file management.

Disadvantages

- Traditional FTP sends usernames and passwords in plain text.
- Less secure than modern alternatives.
- Separate control and data connections add complexity.

Secure Alternatives

- **FTPS (FTP Secure)** – FTP with SSL/TLS encryption.
- **SFTP (SSH File Transfer Protocol)** – Runs over SSH and is much more secure.

RSA Algorithm

RSA (Rivest–Shamir–Adleman) is an **asymmetric (public-key) cryptographic algorithm** used for:

- Encryption and decryption
- Digital signatures
- Key exchange

RSA uses **two keys**:

1. **Public Key** → Used for encryption
2. **Private Key** → Used for decryption

RSA Working

RSA has 3 phases:

1. Key Generation

2. Encryption

3. Decryption

Step 1: Key Generation

Choose two prime numbers

$$\begin{aligned} p &= 3 \\ q &= 11 \end{aligned}$$

Calculate n

$$\begin{aligned} n &= p \times q \\ n &= 3 \times 11 = 33 \end{aligned}$$

Calculate Euler's Totient Function

$$\begin{aligned} \phi(n) &= (p - 1)(q - 1) \\ \phi(n) &= (3 - 1)(11 - 1) = 20 \end{aligned}$$

Choose Public Key e

Choose a number such that:

$$\begin{aligned}1 &< e < \phi(n) \\ \gcd(e, \phi(n)) &= 1\end{aligned}$$

Choose:

$$e = 3$$

Calculate Private Key d

Find d such that:

$$\begin{aligned}d \times e &\equiv 1 \pmod{\phi(n)} \\ d \times 3 &\equiv 1 \pmod{20} \\ d &= 7\end{aligned}$$

because:

$$7 \times 3 = 21 \equiv 1 \pmod{20}$$

Keys Obtained

Public Key

$$(e, n) = (3, 33)$$

Private Key

$$(d, n) = (7, 33)$$

Step 2: Encryption

Suppose the message:

$$M = 2$$

Encryption formula:

$$\begin{aligned}C &= M^e \bmod n \\ C &= 2^3 \bmod 33 \\ C &= 8\end{aligned}$$

Ciphertext:

$$C = 8$$

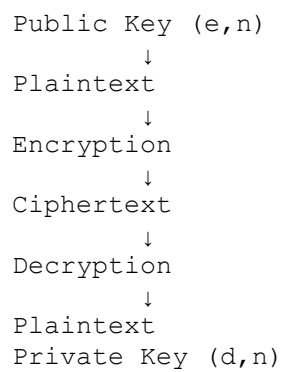
Step 3: Decryption

Decryption formula:

$$\begin{aligned}M &= C^d \bmod n \\M &= 8^7 \bmod 33 \\M &= 2\end{aligned}$$

Original message recovered.

RSA Diagram



Why RSA is Secure

RSA security depends on the difficulty of **factorizing large numbers**.

Given:

$$n = p \times q$$

Finding p and q from a very large n is computationally difficult.

Real RSA uses primes hundreds or thousands of bits long.

Telnet

Telnet (TELEcommunication NETwork) is an **Application Layer protocol** that allows a user to **remotely access and control another computer** over a network.

It provides a **command-line interface** to a remote machine.

Purpose of Telnet

Using Telnet, a user can:

- Log in to a remote computer
- Execute commands remotely
- Manage servers and network devices

```
Client PC
  ↓
Telnet
  ↓
Remote Server
```

How Telnet Works

1. User starts a Telnet session.
2. Connection is established with the remote host.
3. User enters username and password.
4. Commands typed on the local machine are executed on the remote machine.
5. Results are displayed on the user's screen.

```
User
  ↓
Telnet Client
  ↓
Internet
  ↓
Telnet Server
```

Port Number

Telnet uses:

TCP Port 23

Example

```
telnet 192.168.1.10
```

or

```
telnet example.com
```

This attempts to connect to the remote system.

Advantages

- Simple remote access.
- Easy server administration.
- Useful for testing network services.

Disadvantages

The biggest problem:

✗ No encryption

Telnet sends:

- Username
- Password
- Commands
- Data

in **plain text**.

An attacker can capture everything using packet sniffing.

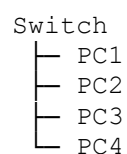
VLAN (Virtual Local Area Network)

A **VLAN (Virtual Local Area Network)** is a logical division of a physical LAN into multiple separate broadcast domains.

It allows devices to be grouped together **logically** even if they are connected to the same physical switch.

Why VLAN is Needed?

Without VLAN:



All devices are in the same broadcast domain.

A broadcast sent by one PC reaches all others.

With VLAN

```
Switch
├─ VLAN 10 → PC1, PC2 (Sales)
└─ VLAN 20 → PC3, PC4 (Accounts)
```

Now:

- VLAN 10 and VLAN 20 are separate networks.
- Broadcasts from VLAN 10 do not reach VLAN 20.
- Improves security and performance.

Example

A company has two departments:

```
Sales Department
Accounts Department
```

Even if all PCs are connected to the same switch, VLANs can separate them:

```
VLAN 10 → Sales
VLAN 20 → Accounts
```

Advantages of VLAN

1. Improved Security

Users in one VLAN cannot directly access another VLAN.

2. Reduced Broadcast Traffic

Broadcasts stay within the VLAN.

3. Better Network Management

Users can be grouped logically rather than physically.

4. Flexibility

Moving a user to another department only requires changing VLAN membership.

HTTP (HyperText Transfer Protocol)

HTTP (HyperText Transfer Protocol) is an **Application Layer protocol** used for communication between a **web browser (client)** and a **web server**.

It is the foundation of the **World Wide Web (WWW)**. Today, most websites use **HTTPS** instead of HTTP.

Purpose of HTTP

HTTP is used to:

- Request web pages
- Transfer HTML documents
- Transfer images, videos, and other web content

```
Browser (Client)
    ↓ HTTP Request
Web Server
    ↑ HTTP Response
```

How HTTP Works

Step 1: Client Request

You enter:

```
www.google.com
```

The browser sends an HTTP request to the web server.

Step 2: Server Response

The server processes the request and sends back the webpage.

```
Client
    ↓ Request
Server
    ↑ Response
```

HTTP Request Methods

Method	Purpose
GET	Retrieve data/web page
POST	Send data to server
PUT	Update data

Method	Purpose
DELETE	Delete data

Example

```
GET /index.html HTTP/1.1
Host: example.com
```

HTTP Port Number

Default HTTP port:

```
Port 80
```

Characteristics of HTTP

1. Connectionless

After sending a response, the connection can be closed.

2. Stateless

- The server does not remember previous requests.
- Each request is treated independently.

HTTP vs HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not Secure	Secure
Encryption	No	Yes (TLS/SSL)
Port	80	443

DQDB (Distributed Queue Dual Bus)

DQDB (Distributed Queue Dual Bus) is a **MAN (Metropolitan Area Network)** protocol standardized as **IEEE 802.6**.

It uses **two unidirectional buses** (dual bus) to allow communication between multiple stations in a metropolitan area.

Structure of DQDB



- **Bus A** carries data in one direction.
- **Bus B** carries data in the opposite direction.
- Each station is connected to both buses.

Hence the name:

- **Distributed Queue** → Stations share access using distributed queues.
- **Dual Bus** → Two buses are used.

Working

1. A station wanting to transmit sends a request.
2. Stations maintain a distributed queue of transmission requests.
3. When its turn arrives, the station sends data on the appropriate bus.
4. The destination receives the data from the bus.

This ensures fair access to the network.

Features

- Uses two one-way buses.
- Suitable for MANs (city-wide networks).
- Supports voice, video, and data traffic.
- Provides fair access through distributed queuing.
- High-speed communication.

Advantages

- Efficient bandwidth utilization.
- Fair access for all stations.
- Supports large metropolitan networks.
- Can handle different types of traffic.

Disadvantages

- Complex implementation.
- Expensive compared to LAN technologies.
- Largely replaced by newer technologies such as Ethernet and fiber-based networks.

Firewall

A **Firewall** is a network security system that **monitors, filters, and controls incoming and outgoing network traffic** based on predefined security rules.

It acts as a barrier between a trusted internal network and untrusted networks such as the Internet.

Why is a Firewall Needed?

Without a firewall:

Internet → Computer

Anyone may attempt unauthorized access.

With a firewall:

Internet → Firewall → Computer

The firewall checks traffic and blocks suspicious or unauthorized connections.

How a Firewall Works

When a packet arrives:

1. Firewall examines the packet.
2. Compares it with security rules.
3. Decides whether to:
 - Allow the packet
 - Block the packet

```
Incoming Packet
      ↓
    Firewall
   ↙      ↘
Allow    Block
```

Types of Firewalls

1. Packet Filtering Firewall

- Checks source IP, destination IP, port numbers, and protocol.
- Allows or blocks packets based on rules.

2. Stateful Inspection Firewall

- Tracks active connections.

- Makes decisions based on the connection state.
- More secure than simple packet filtering.

3. Application-Level Firewall (Proxy Firewall)

- Examines application data (HTTP, FTP, SMTP, etc.).
- Can block malicious application traffic.

4. Next-Generation Firewall (NGFW)

- Includes deep packet inspection, intrusion prevention, malware detection, etc.

Advantages

- Prevents unauthorized access.
- Protects against many network attacks.
- Controls network traffic.
- Improves overall security.

Limitations

- Cannot stop all attacks.
- Cannot protect against threats already inside the network.
- Incorrect firewall rules can block legitimate traffic.

Applications

- Home networks
- Corporate networks
- Data centers
- Cloud environments

Modern Topics

ISDN (Integrated Services Digital Network)

ISDN (Integrated Services Digital Network) is a set of communication standards that allows the transmission of **voice, data, text, images, and video** over traditional telephone lines in **digital form**.

It was developed to replace the older analog telephone system.

Why ISDN Was Introduced

Traditional telephone networks were analog:

Voice → Analog Signal

ISDN converts communication into digital form:

Voice/Data/Video
↓
Digital

This provides better quality and faster data transmission.

Features of ISDN

- Digital communication
- Supports voice and data simultaneously
- Better quality than analog systems
- Faster connection setup
- Uses existing telephone lines

ISDN Channels

1. B Channel (Bearer Channel)

- Carries voice, data, or video.
- Speed = **64 kbps**

2. D Channel (Delta Channel)

- Used for signaling and control information.
- Speed = **16 kbps or 64 kbps**

Types of ISDN

1. BRI (Basic Rate Interface)

Contains:

$2B + 1D$

- $2 B \text{ channels} \times 64 \text{ kbps} = 128 \text{ kbps}$
- $1 D \text{ channel} \times 16 \text{ kbps}$

Total $\approx$ **144 kbps**

2. PRI (Primary Rate Interface)

In most countries:

$23B + 1D$

(Uses T1 line)

Total speed:

1.544 Mbps

Applications

- Telephone services
- Video conferencing
- Internet access (historically)
- Data transfer between offices

WiMAX (Worldwide Interoperability for Microwave Access)

WiMAX is a wireless broadband communication technology based on the **IEEE 802.16** standard. It provides high-speed Internet access over long distances.

It was designed to offer broadband connectivity to homes, businesses, and mobile users without requiring physical cables.

Working of WiMAX

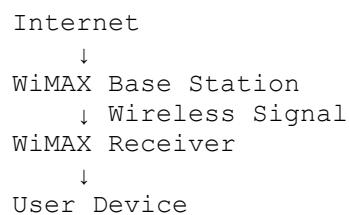
WiMAX consists of:

1. WiMAX Base Station

- Similar to a mobile tower.
- Transmits and receives wireless signals.

2. WiMAX Receiver

- Installed at the user's location.
- Receives Internet signals from the base station.



Features

- High-speed Internet access
- Long-range communication
- Supports fixed and mobile users
- Wireless broadband technology
- Covers large geographical areas

Range and Speed

- Range: Up to **50 km** (ideal conditions)
- Speed: Up to **70 Mbps** (theoretical, varies in practice)

Satellite Transmission

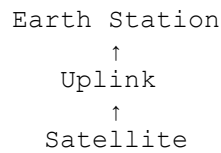
Satellite Transmission is a communication method in which a satellite is used as a relay station to receive signals from one Earth station and retransmit them to another Earth station.

It enables long-distance communication over very large geographical areas.

Working of Satellite Transmission

Step 1: Uplink

The transmitting Earth station sends signals to the satellite.

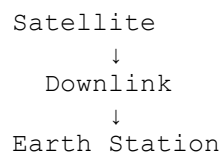


Step 2: Satellite Processing

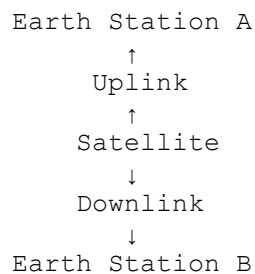
The satellite receives the signal, amplifies it, changes its frequency, and retransmits it.

Step 3: Downlink

The satellite sends the signal back to the receiving Earth station.



Complete Diagram



Components

1. **Earth Station** – Sends and receives signals.
2. **Satellite (Transponder)** – Receives, amplifies, and retransmits signals.
3. **Uplink** – Transmission from Earth to satellite.
4. **Downlink** – Transmission from satellite to Earth.